

Lesson 1

Definitions

What is statistics ?

Statistics is a suit of tools for collecting, summarizing, visualizing and analyzing data. It involves design, description and inferences.

What does the field of statistics have to offer ?

Statistics provides a suit of tools for collecting, summarizing, visualizing and analyzing data. It involves design, description and inferences.

Population

The total set of subjects of interest in the study.

Ex: Male students in the University system in Sri Lanka.

Sample

representative subset of population.
(A subset of the population on which the study collects data)

Ex: Male students at Rajarata University.

Random sampling

Random sampling

- with replacement
 - * is used to find probability with replacement.
- without replacement
 - * is way to find probability without replacement

Random sampling is a part of sampling technique in which each sample has an equal probability of being chosen.

Variable

A measurement of interest that varies between individuals of a population.

Variable

Numerical (Numbers) Categorical (Categories)

Continuous

* can assume any value

Ex: height
Weight

Discrete

* whole numbers

Ex: members of a family
no. of fingers

nominal

* Can't be measured

Ex: colours

ordinal

* Can be ordered

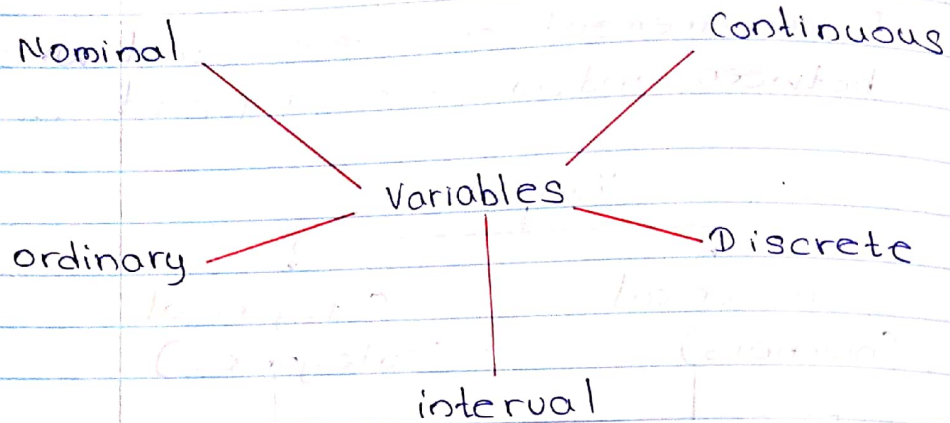
Ex: letter grade for an exam
CA, B, B⁺

Interval

* Constant intervals

Ex: 40kg - 50kg

no of people get
A and B



Summarizing data

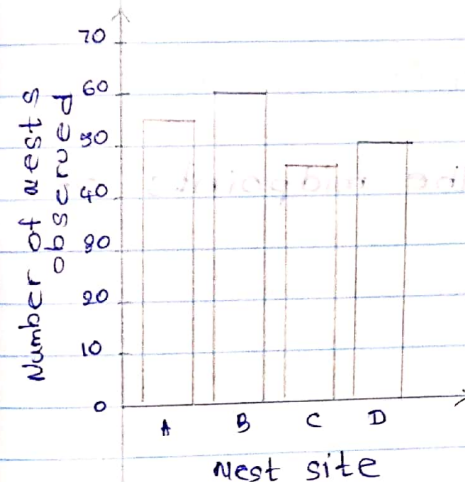
Frequency

listing of all the observed values of particular variable and how many times each value is observed.

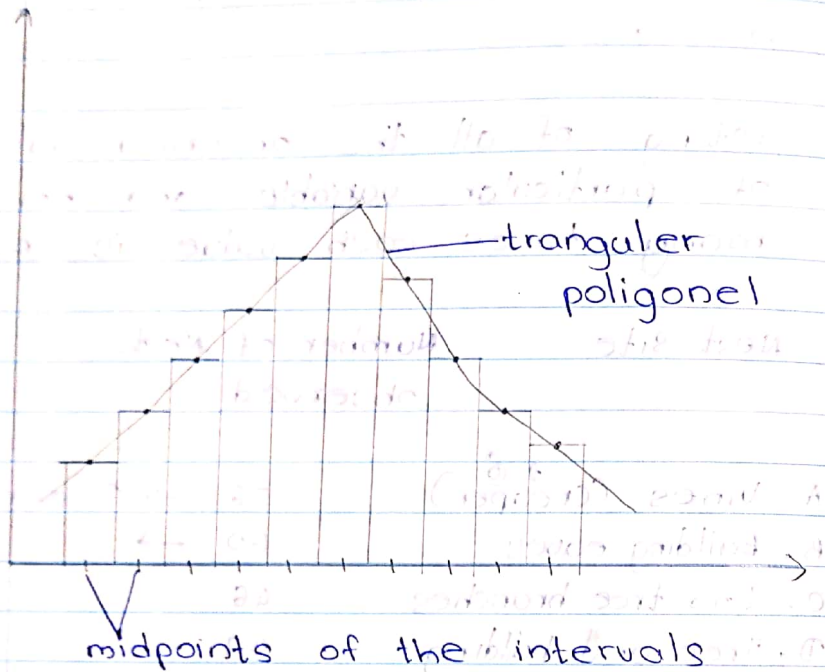
Nest site	Number of Nests observed
-----------	--------------------------

A. Vines (Creeper)	56 → Discrete
B. Building eaves	60 →
C. Low tree branches	46
D. Tree and building cavities	49

Bar chart



Histogram



* Columns are attached.

* Continuous.

Frequency polygon

by connecting the midpoints of histogram.

Relative frequency

$$R.f = \frac{f}{\Sigma f}$$

total frequency

addition or summation.

Phosphorous (mg/g of leaf)	Frequency (i.e. no of determination)	Cumulative frequency	
		Starting with low value	Starting with high value
8.15 - 8.25	2	2	130
8.25 - 8.35	6	8	128
8.35 - 8.45	8	16	122
8.45 - 8.55	11	27	114
8.55 - 8.65	17	44	103
8.65 - 8.75	17	61	86
8.75 - 8.85	24	85	69
8.85 - 8.95	18	103	45
8.95 - 9.05	13	116	27
9.05 - 9.15	10	126	14
9.15 - 9.25	4	130	4

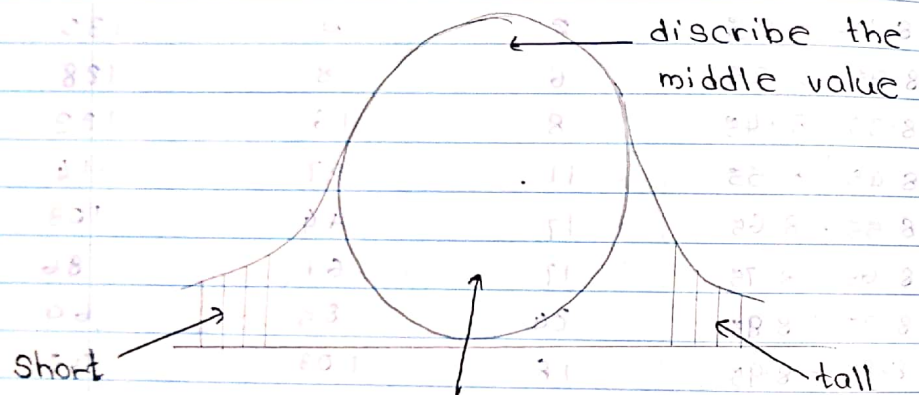
$$\text{Total frequency} = 130 = n$$

$$\Sigma R.f = \frac{2}{130} + \frac{6}{130} + \dots$$

$$= 1 //$$

Measures of central tendency

In a sample of a population, most data points assume values somewhere around the middle of the range of observed values. The concentration of values near the middle is described using measures of central tendency.



majority in the middle
concentration of values
near this middle is
described by measure
of central tendency.

i) Mean

$$\text{Population mean} = \frac{\sum_{i=1}^n x_i}{N}$$

N = Population size

$$\text{Sample mean} = \frac{\sum_{i=1}^n x_i}{n}$$

n = no. of elements

3.3, 3.5, 3.6, 3.6, 3.7, 3.8, 3.8...

$$\sum_{i=1}^8 x_i = 3.3 + 3.5 + 3.6 + 3.6 + 3.7 + 3.8 + 3.8 + 3.8$$

$$\sum_{i=1}^5 x_i = 3.3 + 3.5 + 3.6 + 3.7 + 3.8$$

$$\sum x = \text{entire total} = 95$$

$$\bar{x} = \frac{\sum x_i}{n} = \frac{95}{24} = 3.96 \text{ cm}$$

$$x = 2, 9, 3, 9, 4, 25$$

$$\bar{x} = \frac{40}{6} = 6.67 \text{ cm}$$

$$x = 0, 0, 25, 30, 30, 40$$

$$\Sigma x = 125$$

$$\bar{x} = \frac{125}{6} = 20.83 \text{ cm}$$

mean is influenced by extreme value.

Mean doesn't tell us exact data distribution, because it influence by extreme value.

Extreme value - a value which outside general trend of data. (Outlier)

b) Median

Middle value in an ordered subset.

ascending or
deccending

$$M.D = \frac{X(n+1)}{2}$$

$$\frac{(n+1)}{2} \rightarrow \text{term } (1, 2, 3, \dots)$$

1, 2, 4, 6, 8
↓
median

$$\frac{X(n+1)}{2} = \frac{5+1}{2} = 3^{\text{rd}} \text{ term}$$

1, 2, 4, 6, 8, 11
↓
median

$$\frac{X(n+1)}{2} = \frac{(6+1)}{2} = 3.5^{\text{th}} \text{ term}$$

$$\frac{4+6}{2} = 5 //$$

ES Allen

c) Mode

Mode is the most frequently occurring measurement / element.

1, 2, 3, 3, 3, 4, 5

mode $\rightarrow 3$

Measures of Dispersion

Information about the spread of data. How data spread.

Range

Range = Highest value - Lowest value

Brand A

Brand B

10

85

60

45

50

30

30

85

40

40

20

25

is count when finding the range

$$\text{Range (A)} = 60 - 10 = 50$$

$$\text{Range (B)} = 45 - 25 = 20$$

A is higher range than B.

sample A has a greater variation in drying times in comparison to sample B

Variance

$$\text{Population Variance } \sigma^2 = \frac{\sum (x - \mu)^2}{n}$$

Population Size

$$\text{Sample Variance } s^2 = \frac{\sum (x - \bar{x})^2}{(n-1)}$$

Sample mean
degree of freedom

Degree of freedom mean the no of elements then can change.

	X_i
1	2
2	1
3	9
4	2
5	2

$$\Sigma x_i = 10$$

Degree of freedom = $5 - 1 = 4$
freedom

Point A

x	$x - \mu$	$(x - \mu)^2$
10	-25	625
60	+25	625
50	+15	225
30	-5	25
40	+5	25
20	-15	225
$\Sigma 10$	0	1750

$$\Sigma x_A = 210$$

$$\mu = \frac{210}{6}$$

$$\sigma^2 = \frac{\Sigma (x - \mu)^2}{n}$$

$$= \frac{1750}{6}$$

$$= 291.67 \text{ min}^2$$

Point B

x	$x - \mu$	$(x - \mu)^2$
35	0	0
45	10	100
30	-5	25
35	0	0
40	5	25
25	-10	100
$\Sigma 210$	0	$\Sigma = 250$

$$\Sigma x_B =$$

$$\mu = \frac{210}{6} = 35$$

$$\sigma^2 = \frac{\Sigma (x - \mu)^2}{n}$$

$$= \frac{250}{6}$$

$$= 41.67 \text{ min}^2$$

No: _____ Date: ____/____/____
Paint A has higher variance in comparison to paint B.

standard deviation

$$\text{Population standard deviation } (\sigma) = \sqrt{\sigma^2}$$

$$\text{Sample standard deviation } (s) = \sqrt{s^2}$$

square root of variance.

$$\sigma_A = \sqrt{291.67} \\ = 17.2 \text{ min}$$

$$\sigma_B = \sqrt{41.67} \\ = 6.5 \text{ min}$$

Which is the better paint?
Paint B

because as shown by range, variance and standard deviation, the variation in drying times

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of batches of paints in paint B is lower than paint A.

05.05.2021
Coefficient of variation

$$\text{Population Cvar} = \frac{\sigma}{\mu} \times 100\%$$

Population standard deviation
population mean

$$\text{Sample Cvar} = \frac{s}{\bar{x}} \times 100\%$$

sample standard deviation
sample mean

reason for calculating coefficient of variation is to compare standard deviations.

(from paint question)

$$\sigma_A = 17.1 \text{ min}$$

$$\sigma_B = 6.5 \text{ min}$$

$$CVA = \frac{\sigma}{\mu} \times 100\%$$

$$= \frac{17.1}{35} \times 100\%$$

$$= 48.9\%$$

$$CV_B = \frac{6.5}{35} \times 100\%$$

$$= 18.6\%$$

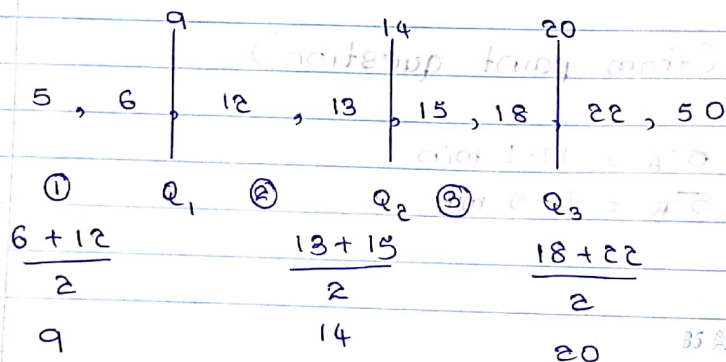
Quartile

Quarter $\rightarrow \frac{1}{4}$ equal parts

$Q_1 = 9$

$Q_2 = 14$ median of data

$Q_3 = 20$ third quartile



Interquartile range.

difference between Q_1 and Q_3

$$I.R = Q_3 - Q_1$$

$$= 20 - 9$$

$$= 11$$

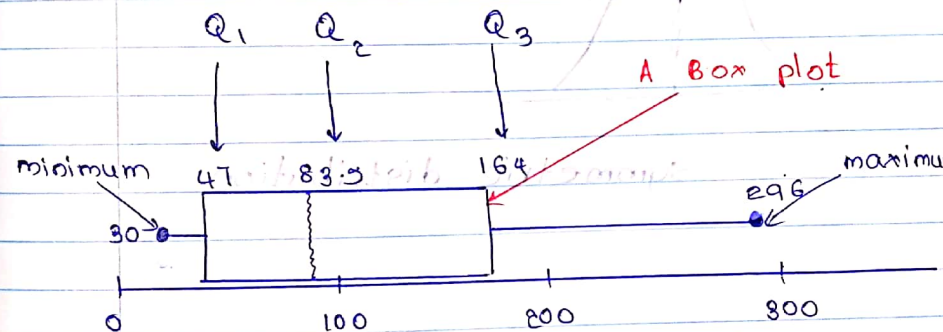
30, 39, 47, 48, 78, 89, 138, 164, 215, 296

$$\frac{(n+1)}{2} = \frac{10+1}{2} = 5.5^{\text{th}} \text{ term}$$

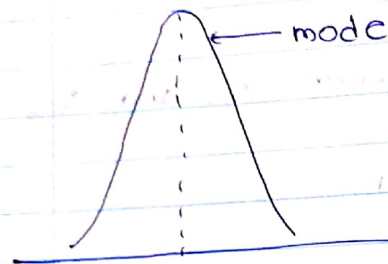
$$\text{median} = \frac{78+89}{2} = 83.5$$

$$Q_2 = 47$$

$$Q_3 = 164$$

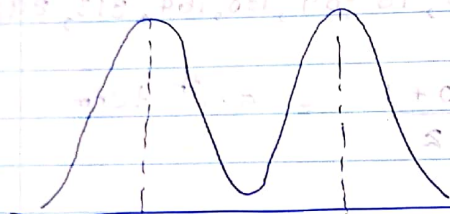


Visualizing the spread of data



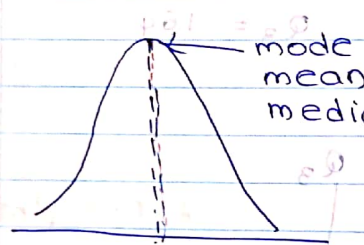
Unimodal

1, 2, (3, 3, 3), 6, 7, 8, 5



Bimodal

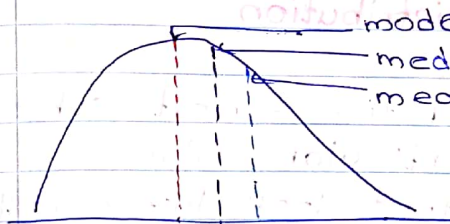
1, 2, (3, 3, 3), 4, (5, 5, 5)



mode
mean
median

all three are in middle

Symmetric distribution



mode

median

mean

Positively skewed

note position of mode, median and mean.

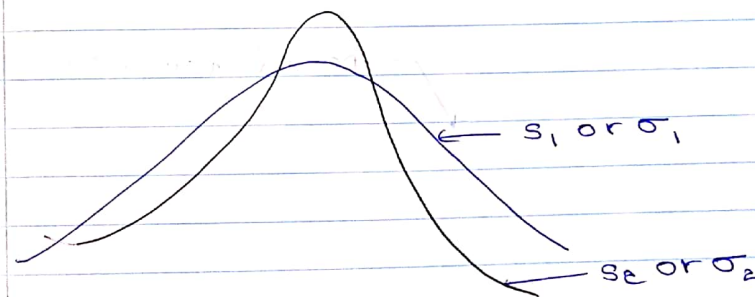


mean

median

mode

negatively skewed



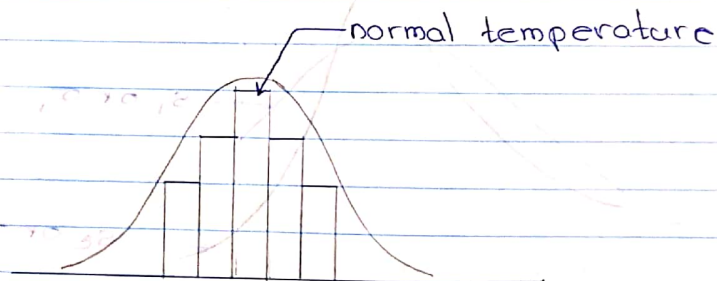
$s_1 > s_2$

because s_1 has greater spread. data can found all over.

Normal Distribution

- Normal distribution - most subject have a particular value.
- Distribution that look like a bell shape is known as normal distribution.
- Normal distribution variables are continuous.
- Normal distribution introduced by Carl Friedrich Gauss.
- Most biological variables are normally distributed.

Ex: Height
Weight
Blood sugar.

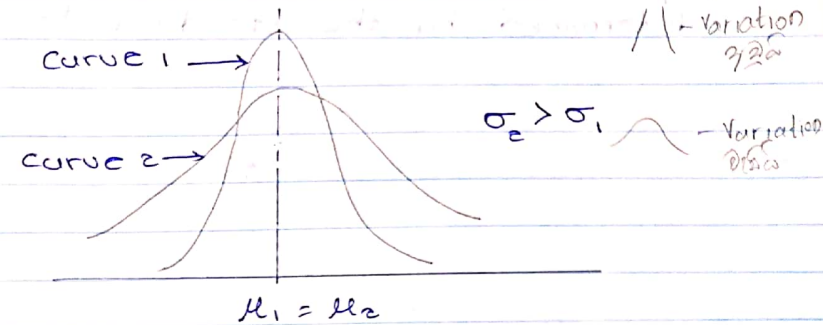


$$y = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

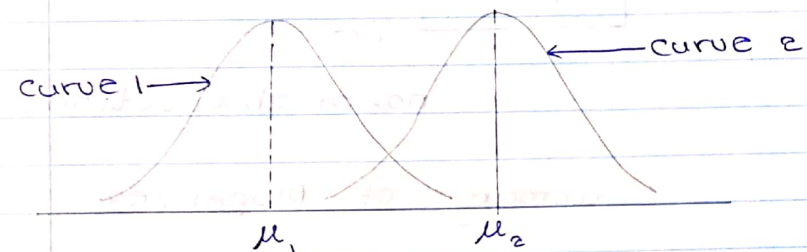
not hard to remember

There are two parameters influence the shape

1. Population mean
2. Standard deviation



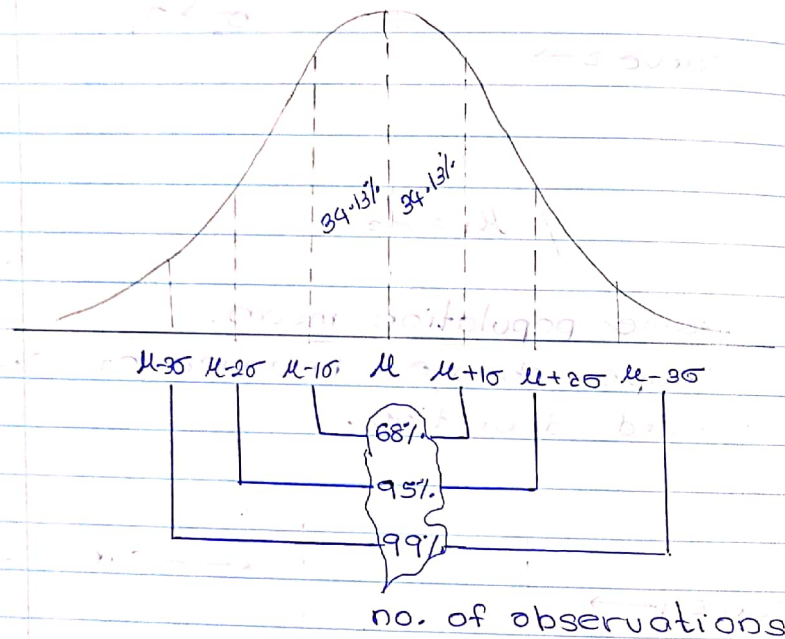
same population mean.
shape changes because of standard deviation.



Mean is different.
The position is influenced by the mean.
Same standard deviation.

In a normal distribution shape is determined by the standard deviation.

In the normal distribution shape is determined by the standard deviation.



Summary of Properties

01. A normal distribution is bell shaped.
02. Unimodal.
03. Symmetrical.
04. The curve never touch the x axis.
05. The total area under the curve is 100% or 1.

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06. Curve is continuous. no gaps.
07. In normal distribution mean mode and median are equal, and located in the center.
08. Area under the curve that lies within 1 standard deviation of the mean is approximately 68%, two standard deviation 95%, three standard deviation 99%.

Different variable have differently shaped and differently positioned normal curves.

Because variables have different means and different standard variations. Each variable have its own curve. There for it is difficult to compare.

Standard normal distribution.

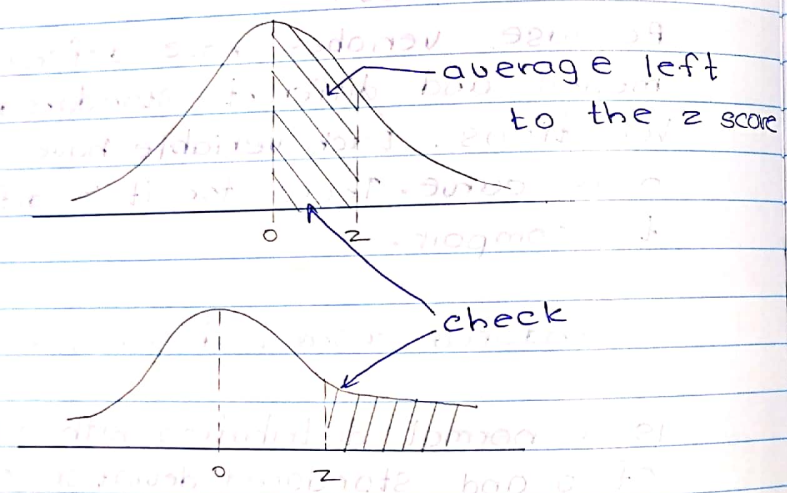
is a normal distribution with mean of 0 and standard deviation of 1.

We can convert any normal distribution variable to standard normal distribution format using below equation.

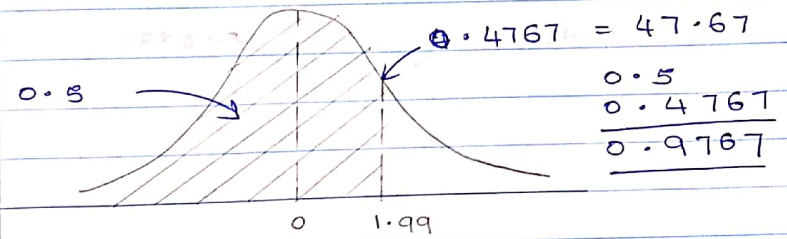
15 Alice

$$z = \frac{x - \mu}{\sigma}$$

$x - \mu$ = Different between variable and population mean.
 μ = Population mean.
 σ = Population standard deviation.
 σ (standardizing by σ).



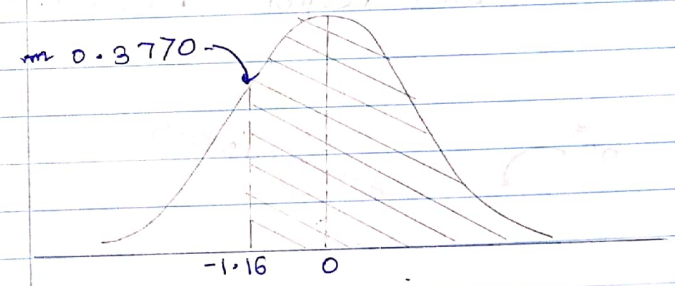
Q01. Find the area to the left of, $z = 1.99$.



$$= 0.5 - 0.4767$$

$$= 0.0233 //$$

Q02. Find the area to the right of $z = -1.16$.



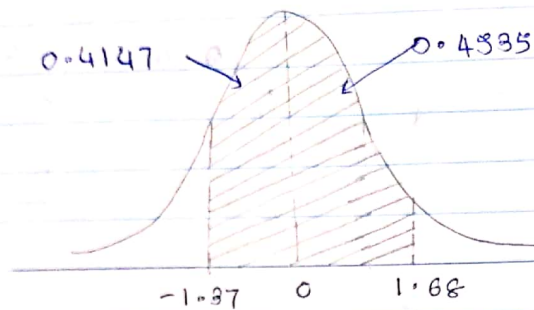
$$0.3770$$

$$0.5000$$

$$0.8770$$

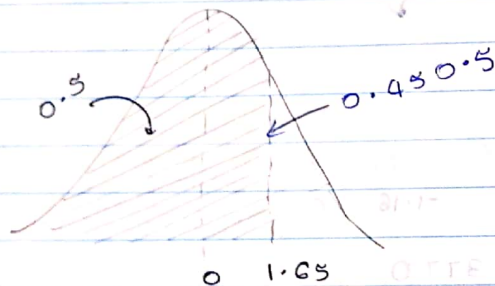
$$87.7 \%$$

03. Find the area between $z = 1.68$ and $z = -1.37$.



$$\begin{array}{r} 0.4535 \\ 0.4147 \\ \hline 0.8682 \end{array} \quad 86.82\%$$

04. Probability that a variable can be found in this region $P(z < 1.65)$



$$\begin{array}{r} 0.4505 \\ 0.5000 \\ \hline 0.9505 \end{array} \quad \begin{array}{l} \text{H.W. } 1.65 \\ P(0 < z < 1.65) \\ P(z < 1.65) \end{array}$$

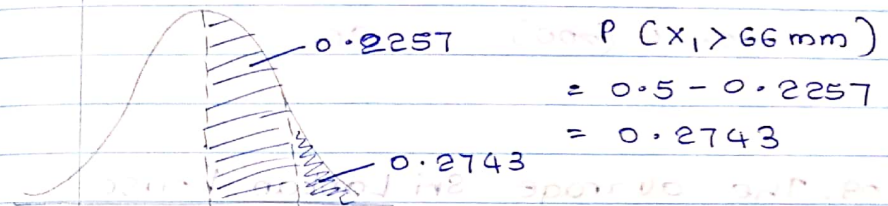
Bone length $\mu = 60 \text{ mm}$, $\sigma = 10 \text{ mm}$.

1) What is the population of bone lengths larger than 66 mm?

$$z = \frac{x - \mu}{\sigma}$$

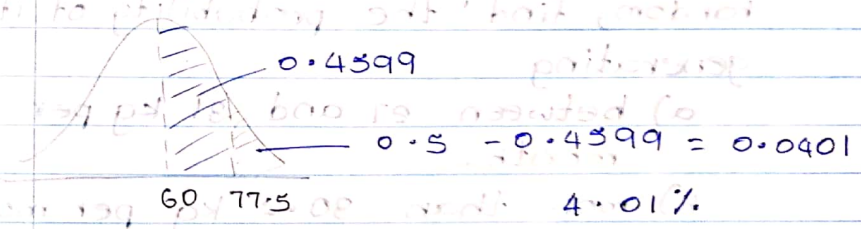
$$z = \frac{66 - 60}{10}$$

$$= 0.60$$



$$\begin{array}{l} P(X_1 > 66 \text{ mm}) \\ = 0.5 - 0.2257 \\ = 0.2743 \end{array}$$

2) What is the proportion under the value that lies to the right of 77.5 mm?



$$\begin{array}{l} z = \frac{77.5 - 60}{10} \\ = 1.75 \end{array}$$

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If there are 2000 bone lengths how many are larger than 77.5mm.

$$(0.0407)(2000) = 80$$

40% got A grades. If there 200 students, how many got A passes.

$$\frac{40}{100} \times 200 = 80$$

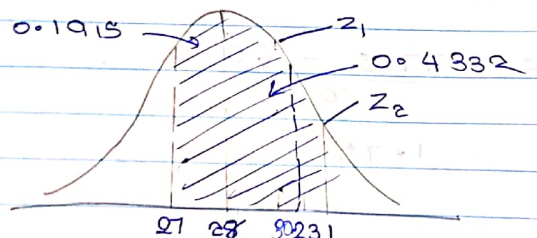
$$(0.40)(200) = 80$$

03. The average Sri Lankan house hold generates 28kg of news paper each month. Assume that the standard deviation 2kg. If a house hold is selected at random, find the probability of its generating.

a) between 27 and 31 kg per month.

b) more than 30.2 kg per month.

a)



No:

Date: / /

$$z_1 = \frac{x_1 - \mu}{\sigma} \quad z_2 = \frac{x_2 - \mu}{\sigma}$$

$$z_1 = \frac{27 - 28}{2} = -0.5 \quad z_2 = \frac{31 - 28}{2} = 1.5$$

$$z_1 = -0.5 \quad z_2 = 1.5$$

$$0.1915$$

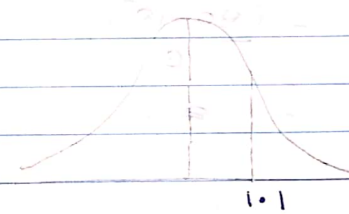
$$0.4332$$

$$0.6247$$

$$b) \quad z_3 = \frac{x_3 - \mu}{\sigma}$$

$$= \frac{30.2 - 28}{2}$$

$$= \frac{2.2}{2} = 1.1$$



$$0.5000$$

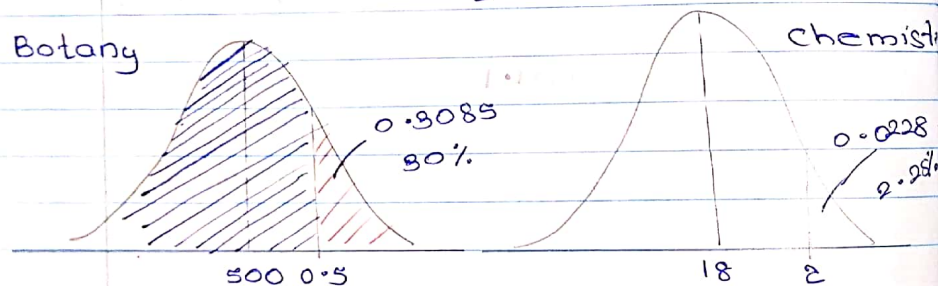
$$0.8643$$

$$0.3643$$

04. You took a Botany test 550. The population that took botany has a $\mu = 500$ and $\sigma = 100$. Your friend took a chemistry test and scored 80. The population that took chemistry has $\mu = 18$ and a $\sigma = 6$. who did better?

$$\begin{aligned}\text{Botany} \rightarrow z_1 &= \frac{(x - \mu)}{\sigma} \\ &= \frac{550 - 500}{100} \\ &= 0.5\end{aligned}$$

$$\begin{aligned}\text{Chemistry} \rightarrow z_2 &= \frac{(x - \mu)}{\sigma} \\ &= \frac{(80 - 18)}{6} \\ &= 10.33\end{aligned}$$



friend(Chemistry) is better.

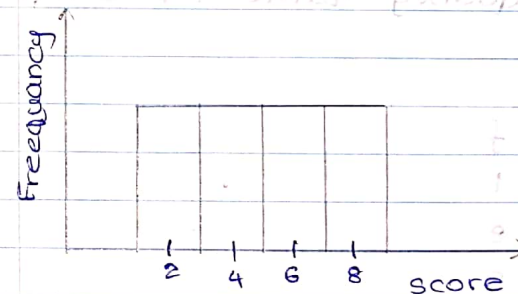
Q12. has 4 student. They get a score of 2, 6, 4, 8 out of 8 points. calculate μ and σ .

4 $\rightarrow$ Population

$$\mu = \frac{2 + 6 + 4 + 8}{4} = 5$$

$$\begin{aligned}\sigma &= \sqrt{\frac{(2-5)^2 + (6-5)^2 + (4-5)^2 + (8-5)^2}{4}} \\ &= 2.236\end{aligned}$$

draw a histogram for this.



* uniform distribution

Take a sample size of 100 with replacement. calculate mean for each sample.

sample	mean	sample	mean
2, 2 →	2 < P.mean	6, 2 →	4
2, 4 →	3	6, 4 →	5
2, 6 →	4	6, 6 →	6
2, 8 →	5 = P.mean	6, 8 →	7
4, 2 →	3	8, 2 →	5
4, 4 →	4	8, 4 →	6
4, 6 →	5	8, 6 →	7
4, 8 →	6 > P.mean	8, 8 →	8

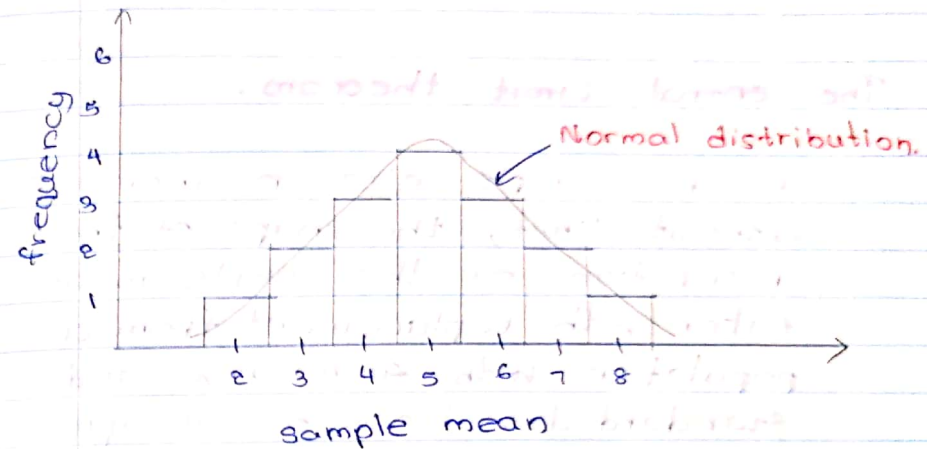
We always don't get the population mean.

⇒ sampling error.

create frequency table for sample mean.

$\bar{x}$	f
2	1
3	2
4	3
5	4
6	3
7	2
8	1

construct histogram for $\bar{x}$ and f.



calculate $\mu_{\bar{x}}$ and $\sigma_{\bar{x}}$.
 The mean of sample means standard deviation of sample means

$$\mu_{\bar{x}} = \frac{2 + 3 + 4 + 5 + 4 + 3 + 6 + 7 + 5 + 6 + 7 + 8}{16}$$

$$= 5 \leftarrow \text{close to our population mean.}$$

$$\sigma_{\bar{x}} = \sqrt{\frac{(2-5)^2 + (3-5)^2 + (4-5)^2 + \dots}{16}}$$

$$= 1.581$$

standard error of the mean

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{2.536}{\sqrt{2}} = 1.581$$

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$$\mu = \mu_{\bar{x}} \quad \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

The Central Limit theorem.

- as the sample size n increases without limit, the shape of the distribution of the sample means taken with replacement from a population with mean μ and standard deviation σ will approach a normal distribution. As shown this distribution mean μ and standard deviation $\sigma/\sqrt{n}$

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

The central limit theorem is applicable only when,

1. The original variable is normally distributed, the distribution
2. of the sample means will be for any sample size n normally distributed, for any sample size n .

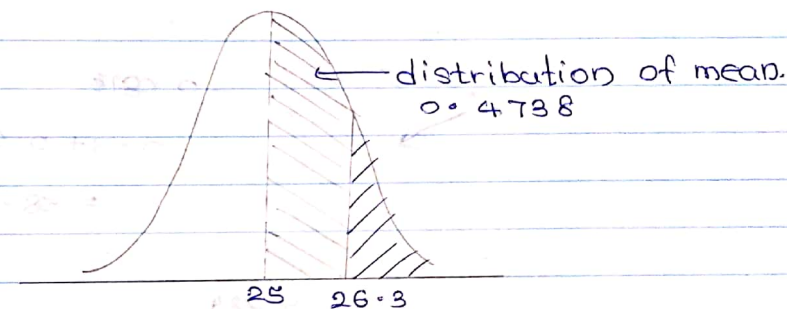
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2. when the distribution of the original variable might not be normal, a sample size of 30 or more is needed to use a normal distribution to approximate the distribution of the sample means.

02/06/2021

A survey of children between ages of 2 and 5 found they watched an average of 25 hours of T.V per week. Assume the variable is normally distributed with a standard distribution of 3 hours. If 20 children are selected at random. Find the probability that the mean^{no} of hours that they watched television will be greater than 26.3 hours.



35/11/20

$$z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} = \frac{26.8 - 25}{3 / \sqrt{20}} = 0.671$$

$$= 1.94$$

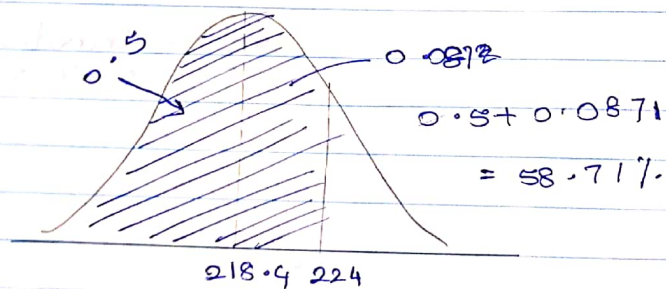
$$0.5 - 0.4738 = 0.0262$$

$$2.62\%$$

$$P(\bar{x} > 26.8) = 2.62\%$$

The average number of kg of vegetable consumed by a person for year 218.4 kg. Assume that standard deviation is 25 kg and the distribution is normal.

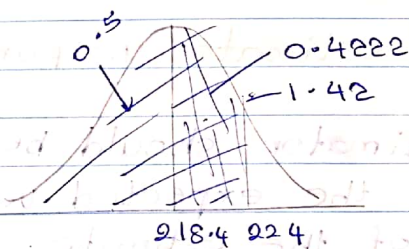
a) Find the probability that a person selected at random consumes less than 224 kg/yr



$$z = \frac{x - \mu}{\sigma} = \frac{224 - 218.4}{25} = 0.22$$

$$= 0.22$$

b) If a sample of 40 individuals is selected, find the probability that the mean of the sample will be less than 224 kg per year.

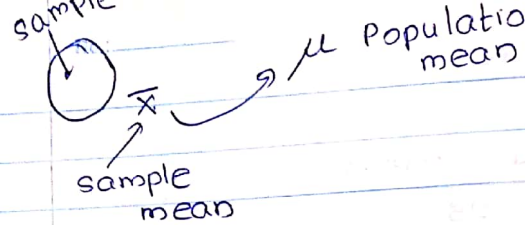


$$z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} = \frac{224 - 218.4}{25 / \sqrt{40}} = 1.42$$

$$= 1.42$$

$$0.9222$$

$$92.22\%$$



A point estimate →

Is a specific numerical value estimate of an parameter. The best point estimate of μ is the sample mean $\bar{x}$.

Good estimator properties.

01. The estimator should be unbiased. That is the expected value of the mean of the estimates obtain from the sample of given size is equal to the parameter being estimate.

$$\frac{\bar{x}_1 + \bar{x}_2 + \bar{x}_3 + \bar{x}_4 + \bar{x}_5}{5} = \mu$$

02. The estimator should be consistent. For a consistent estimator, as sample size increases, the value of the estimator approaches the

value of the parameter estimated.

03. The estimator should be relatively efficient.

That is of all the statistic that can be used to estimate parameter relatively efficient estimator has the smallest variance.

Interval estimate

$$\bar{x}_1, \bar{x}_2, \bar{x}_3, \bar{x}_4, \bar{x}_5$$

μ

Interval estimate - Interval which contains μ , is between the mean values of $\bar{x}_1$ and $\bar{x}_4$.

- * Interval estimate of a parameter is an interval or range of value used to estimate the parameter.

Confident Intervals.

The confident interval of an interval estimate of an parameter is the

No: _____

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probability that interval that contain the parameter.

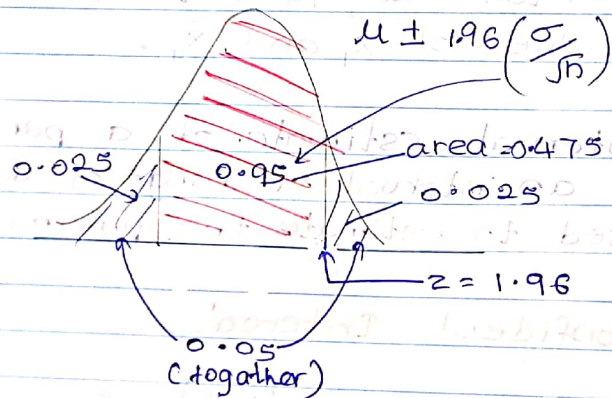
$$\bar{x}_1 - \bar{x}_4 \leftarrow 95\% \text{ confidence level}$$

A confidence interval is a special interval estimator of a parameter determined by using data obtained from a sample and by using the specific confidence level.

In statistics we use two confidence levels

1. 95% 2. 99%

C.I 95% $\rightarrow \mu \pm 1.96 \left(\frac{\sigma}{\sqrt{n}} \right)$



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If a specific sample mean is selected, say $\bar{x}$, there is a 95% chance (or probability) that the interval $\mu \pm 1.96 \left(\frac{\sigma}{\sqrt{n}} \right)$ contain $\bar{x}$.

Like wise, there is a 95% probability that the interval specified by $\bar{x} \pm 1.96 \left(\frac{\sigma}{\sqrt{n}} \right)$ will contain μ .

95% confidence interval for μ :

$$95\% \Rightarrow \bar{x} - 1.96 \left(\frac{\sigma}{\sqrt{n}} \right) < \mu < \bar{x} + 1.96 \left(\frac{\sigma}{\sqrt{n}} \right)$$

95% - C.I

1.96 - interval

Q A survey of 30 adults found that the mean age of a person's car is 5.6 years. assuming that the standard deviation of the deviation of the population is 0.8 years. Find the best point estimate of the μ and the 95% C.I of μ . (The best point estimate is 5.6 years.)

$$\left(5.6 \pm 1.96 \left(\frac{0.8}{\sqrt{30}} \right) \right)$$

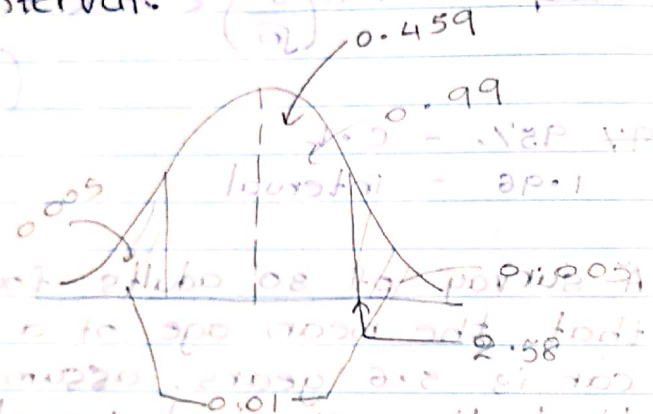
$$\bar{x} - 1.96 \left(\frac{\sigma}{\sqrt{n}} \right) < \mu < \bar{x} + 1.96 \left(\frac{\sigma}{\sqrt{n}} \right)$$

$$5.6 - 1.96 \left(\frac{0.8}{\sqrt{80}} \right) < \mu < 5.8 + 1.96 \left(\frac{0.8}{\sqrt{80}} \right)$$

$$5.3 < \mu < 5.89$$

95% μ will fall between 5.3 and 5.89.

Calculate the 95% confidence interval.



$$\bar{x} - 2.58 \left(\frac{\sigma}{\sqrt{n}} \right) < \mu < \bar{x} + 2.58 \left(\frac{\sigma}{\sqrt{n}} \right)$$

$$5.6 - 2.58 \left(\frac{0.8}{\sqrt{80}} \right) < \mu < 5.8 - 2.58 \left(\frac{0.8}{\sqrt{80}} \right)$$

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Date: / /

$$5.22 < \mu < 5.98$$

$$95\% \rightarrow 5.3 - 5.89 \rightarrow \text{more precise}$$

$$99\% \rightarrow 5.22 - 5.98 \rightarrow \text{Less precise}$$

which is more precise? (range is smaller more precise)

confidence level $\uparrow$ Precision $\downarrow$
sample size $\uparrow$ Precision $\uparrow$

Repeat with sample size of 100.

$$\bar{x} - 1.96 \left(\frac{\sigma}{\sqrt{n}} \right) < \mu < \bar{x} + 1.96 \left(\frac{\sigma}{\sqrt{n}} \right)$$

$$5.6 - 1.96 \left(\frac{0.8}{\sqrt{100}} \right) < \mu < 5.8 + 1.96 \left(\frac{0.8}{\sqrt{100}} \right)$$

$$5.6 - 1.96 (0.08) < \mu < 5.6 + 1.96 (0.08)$$

$$5.4432 < \mu < 5.756$$

$$\bar{x} - 2.58 \left(\frac{0.8}{\sqrt{100}} \right) < \mu < \bar{x} + 2.58 \left(\frac{0.8}{\sqrt{100}} \right)$$

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$$5.6 - 2.58(0.08) < \mu < 5.6 + 2.58(0.08)$$

$$5.39536 < \mu < 5.80464$$

95% CI $\rightarrow 5.44, 5.756$

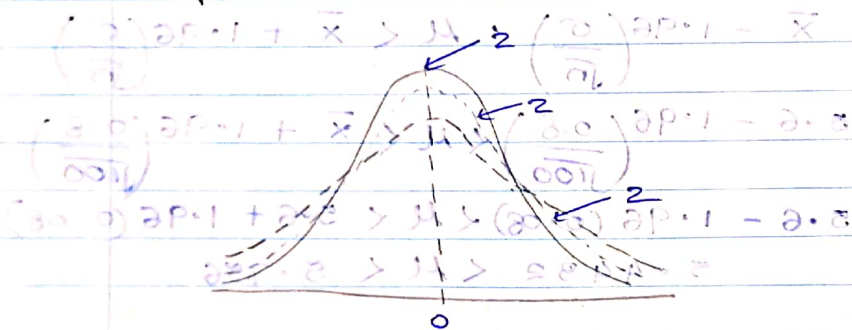
99% CI $\rightarrow 5.39, 5.80$

- * $\sigma \rightarrow$ Population standard deviation.
- * $s \rightarrow$ sample standard deviation.

We can't use standard dev normal distribution. Therefore we use t distribution.

t - distribution.

Properties of t - distribution.



- * It is actually a collection of curves.

No: _____ Similar to 2 (standard)

- * Bell shaped.
- * Symmetric about the mean.
- * Doesn't touch x-axis.
- * Mean, median and mode are equal to zero and are located in the center of the distribute.

Difference from standard.

- * The variance is greater than 1.
- * It is a series of curves that change shape based on the degree of freedom.
- * As sample size increases degree of freedom increases and the t distribution approaches the standard normal distribution.

degree of freedom

$$n - 1 \quad n - \text{sample size}$$

- * degree of freedom increases.

$$\bar{x} - t_{\alpha/2} \left(\frac{s}{\sqrt{n}} \right) < \mu < \bar{x} + t_{\alpha/2} \left(\frac{s}{\sqrt{n}} \right)$$

Q

Ten randomly selected people were asked how long they slept at night. The mean time was 7.1 hours, and the standard deviation was 0.78 hours. Find the 95% confident interval of the mean time.

$$\bar{x} - t_{\alpha/2} \left(\frac{s}{\sqrt{n}} \right) < \mu < \bar{x} + t_{\alpha/2} \left(\frac{s}{\sqrt{n}} \right)$$

$$7.1 - 2.262 \left(\frac{0.78}{\sqrt{10}} \right) < \mu < 7.1 + 2.262 \left(\frac{0.78}{\sqrt{10}} \right)$$

$$7.1 - 0.56 < \mu < 7.1 + 0.56$$

$$6.54 < \mu < 7.66$$

σ was known. sample size was 30 or more of it, the sample size was less than 30, population was normally distributed, $\rightarrow$ z table

With deal with samples, σ is not known. s - sample standard deviation. $\therefore$ we can't use the standard distribution. use $\rightarrow$ t table

The data deals with no of home fires started by candles over a period of six seven years. The mean number of homes is 7041.41 with standard deviation 1610.3 homes. Calculate the 99% C.I for ^{mean} home fires started by candles

$$t = \frac{1}{3} = 3.707$$

$$\bar{x} - t_{\alpha/2} \left(\frac{s}{\sqrt{n}} \right) < \mu < \bar{x} + t_{\alpha/2} \left(\frac{s}{\sqrt{n}} \right)$$

$$7041.4 \pm 3.707 \left(\frac{1610}{\sqrt{7}} \right)$$

$$7041.4 \pm$$

$$4785.2 < \mu < 9297.6$$

is σ known

use $z_{\alpha/2}$ and σ in the formula

Use $t_{\alpha/2}$ value s in the formula

* If $n < 30$ the values must be normally distributed.

1. Find each $z_{\alpha/2}$ for the 99% confidence interval
- $z_{\alpha/2}$ for the 99% confidence interval
 - $z_{\alpha/2}$ for the 98% confidence interval
 - $z_{\alpha/2}$ for the 95% confidence interval
 - $z_{\alpha/2}$ for the 90% confidence interval
 - $z_{\alpha/2}$ for the 94% confidence interval

i. $z_{\alpha/2} = 2.58$

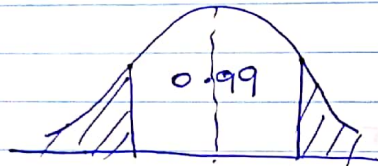
ii. $z_{\alpha/2} = 2.32$

iii. $z_{\alpha/2} = 1.96$

iv. $z_{\alpha/2} = 1.65$

v. $z_{\alpha/2} = 1.88$

i)



$$0.99 \div 2 = 0.495$$

$$z = 2.58$$

21. A sample of the reading scores of 35 fifth graders have a mean of 82. The standard deviation of the population is 15.

- Find the best point estimate of the mean.
- Find the 95% confidence interval of the mean reading scores of all fifth graders.

iii. Find the 99% confidence interval of the mean reading of all fifth grades.

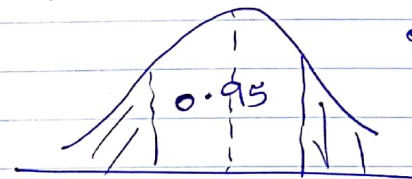
iv. Which interval is larger?

Explain why.

$n = 35$ $\bar{x} = 82$ $\sigma = 15$

i. 82% sample mean

ii.



$$0.475 \rightarrow 1.96$$

$$C.I = \bar{x} \pm 1.96 \left(\frac{\sigma}{\sqrt{n}} \right)$$

$$82 - 1.96 \left(\frac{15}{\sqrt{35}} \right) < \mu < 82 + 1.96 \left(\frac{15}{\sqrt{35}} \right)$$

$$77.03 < \mu < 86.97$$

iii.

$$82 - 2.58 \left(\frac{15}{\sqrt{35}} \right) < \mu < 82 + 2.58 \left(\frac{15}{\sqrt{35}} \right)$$

$$75.45 < \mu < 88.55$$

35

No. _____

Date: ____/____/____

iv. 99% confidence interval for μ
 Larger $\rightarrow$ less precision.

03. Find the value of each,
 i. $t_{\alpha/2}$ and $n=18$ for the 99% C.I for μ

- ii. $n=23$ 95%
- iii. $n=15$ 98%
- iv. $n=10$ 90%
- v. $n=20$ 95%

i. $n-1 = 17$

2.898

ii. 2.074

iii. 2.076

iv. ~~2.624~~ 0.074 (2) 8P.1 $\bar{x} = 20$

v. 1.898 2.074 (1)

04. The average hemoglobin readings for a sample of 20 teachers were 16 grams per 100 milliliters, with a sample standard deviation of 2 grams. Find the 99% confidence interval of the true mean.

$n = 20$ $\bar{x} = 16$ $S = 2$ C.I 99

No. _____

Date: ____/____/____

$$\bar{x} \pm t_{\alpha/2} \left(\frac{s}{\sqrt{n}} \right)$$

$$16 \pm 2.861 \left(\frac{2}{\sqrt{20}} \right)$$

$$16 \pm 1.28$$

Calculating confidence intervals for proportions

Proportions

Ex - % of people who like chocolate cake.

assume - population proportion.

$\hat{p}$ - sample proportion.

x is the number of individuals who possess the feature of interest.

n = sample size.

$$\hat{p} = \frac{x}{n}$$

$1 - \hat{p}$ - Proportion that did not possess the characteristic/feature of interest

No: _____

$$\hat{q} = \frac{n-x}{n}$$

$$\hat{q}' = 1 - \hat{p}$$

In a recent survey of 150 households 54 had air conditioning. Find $\hat{p}$ and $\hat{q}$.

$$\hat{p} = \frac{54}{150} = 0.36$$

$$\hat{p} = \frac{150-54}{150} \text{ or } 1 - \frac{54}{150}$$

Confident intervals

$$\hat{p} - z_{\alpha/2} \sqrt{\frac{\hat{p}\hat{q}}{n}} < p < \hat{p} + z_{\alpha/2} \sqrt{\frac{\hat{p}\hat{q}}{n}}$$

when $n\hat{p} \geq 5$ or $n\hat{q} \geq 5$

No: _____

Date: ____/____/____

A sample of 500 university application included 60 from men. Find the 90% confidence interval of true mean of the true proportion of men who apply to the program.

$$\hat{p} = \frac{60}{500} = 0.12 \quad \hat{q} = 0.88$$

$$z_{\alpha/2} = 1.65$$

$$\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}\hat{q}}{n}}$$

$$0.12 \pm 1.65 \sqrt{\frac{0.12 \times 0.88}{500}}$$

$$0.12 \pm 0.024$$

$$0.096 < p < 0.14$$

90% confident that percentage of applicants who are men is between 9.6% and 14%.

A survey of 1721 people found that 15.9% of the individuals purchased news papers. Find the 95% confident interval of for the true proportion of people who purchase news papers.

$$n = 1721 \quad \hat{p} = 15.9\% = 0.159$$

$$\hat{q} = 0.841 \quad z = 1.96$$

$$\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}\hat{q}}{n}}$$

$$15.9 \pm 1.96 \sqrt{\frac{15.9 \times 0.841}{1721}}$$

$$14.2 < p < 17.6$$

Hypothesis testing

Is a new fertilizer better than is being used right now?

Effects of medication on heart rate?

Such a questions can be addressed through hypothesis testing.

What is hypothesis? Hypotheses - plural

Statistical hypothesis is a conjecture (statement) about a population parameter.

This conjecture or statement can either true or false.

A null hypothesis - H_0

It is a hypothesis that state, that there is no different between parameter and a specific value.

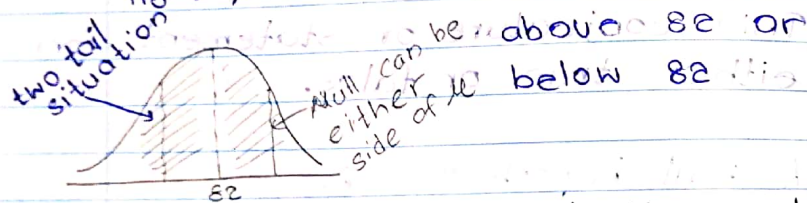
Alternative hypothesis - H_1

Is a state that there is a different between the parameter and specific value.

Ex - A medical researcher is interested in whether a new medication will have any undesirable side effects. The researcher is concerned with the heart rate. Will the medication increase, decrease or remain unchanged after the medication. The researcher knows that the mean heart rate of the population is 82 beats per minute.

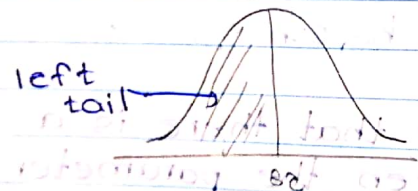
How do we write the H_0 and H_1

$$H_0 : \mu = 82 \quad H_1 : \mu \neq 82$$



Researcher is interested if heart rate is lowered because of medication.

$$H_0 : \mu = 82 \quad H_1 : \mu < 82$$



Researcher is interested in if the heart rate is elevated by medication.

$$H_0 : \mu = 82 \quad H_1 : \mu > 82$$



assume k represent specific number

	two tail	Right tail	Left tail
H_0	$\mu = k$	$\mu = k$	$\mu = k$
H_1	$\mu \neq k$	$\mu > k$	$\mu < k$

A contractor wishes to lower heating bills using a special type of insulation in houses. If the average monthly heating bill is \$78. State the hypotheses about heating constant.

$$H_0 : \mu = 78 \quad H_1 : \mu < 78$$

No: _____

Ho true Ho false

Reject
HoError
type ICorrect
decision← Desirable
outcomeDo not
reject
Hocorrect
decisionError
type II

← not desirable

Type I error - occurs if you reject the null hypothesis when it is true

α - probability of committing type I error

Type II error - occurs if you do not reject the null when it is false.

β - probability

α is easier to compute.

β is difficult to compute.

If $\alpha \uparrow$ then $\beta \downarrow$, $\alpha \downarrow$ then $\beta \uparrow$

α is referred to level of significance

No: _____

Date: ____/____/____

Procedure for carrying out test.

Traditional method.

01. state hypothesis.
02. Find the critical values from the appropriate table.
03. compute test value.
04. Make a decision to reject or not to reject null hypothesis.
05. Summarize results.

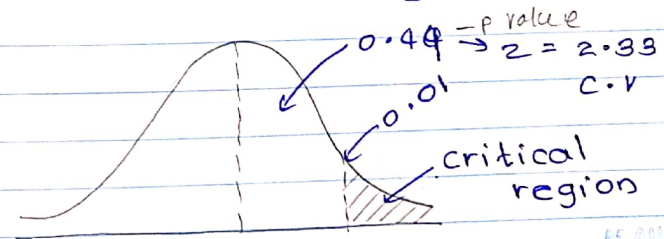
* we testing null, our decision either to accept or ^{fail to} reject null.

Critical Value.

Find the critical value for $\alpha = 0.01$. (right tailed test)

$$H_0: \mu = k, H_1: \mu > k$$

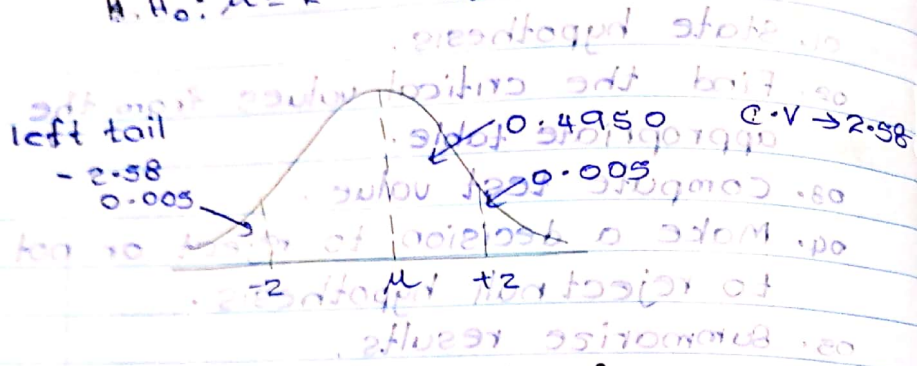
z value corresponding to $\alpha = 0.01$.



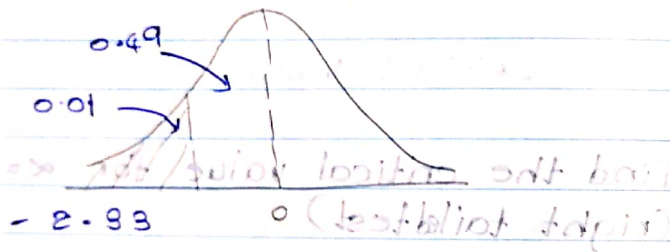
ES ALICE

Find the critical value for $\alpha = 0.01$, for two tailed test.

$$H_0: \mu = k \quad H_1: \mu \neq k$$



Find the critical value for $\alpha = 0.01$ for two tailed. Left tailed.



$$x < \mu, H_0: x = \mu, H_1: x > \mu$$

No:

Date: / /

A researcher reports that the average salary of a teacher is more than Rs 42000. A sample of 30 teachers has a mean salary of Rs 43260. At $\alpha = 0.05$, test the claim that a teacher earns more than Rs 42000 per month. The standard deviation of population $\sigma = 5230$.

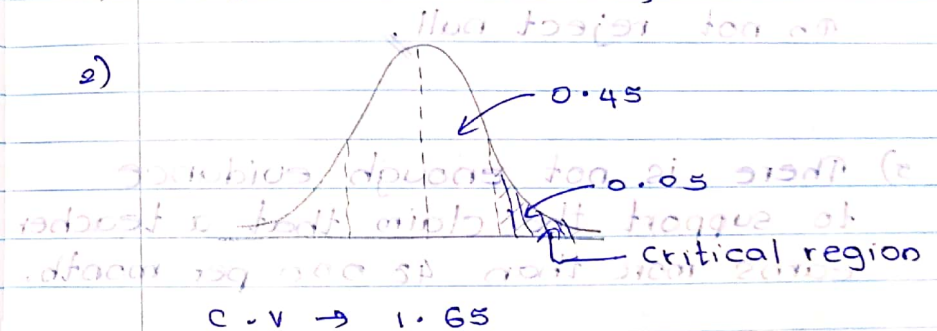
$$H_0: \mu = 42000 \quad H_1: \mu > 42000$$

$$n = 30, \bar{x} = 43260, \alpha = 0.05$$

$$\sigma = 5230$$

$$1) H_0: \mu = 42000 \quad H_1: \mu > 42000$$

2)

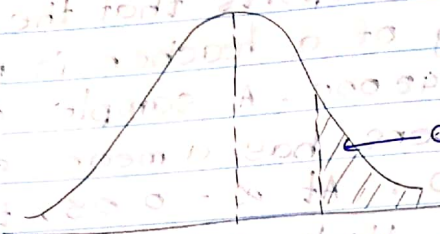


$$3) z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

$$= \frac{43260 - 42000}{5230 / \sqrt{30}}$$

$$= 1.29$$

15/11/20



If test statistic falls in the critical region - Reject null

If test statistic does not fall in the critical region - Do not reject null.

Do not reject null //

- 5) There is not enough evidence to support the claim that a teacher earns more than 42 000 per month.

$$\mu - \bar{x} = 2$$

$$\frac{2}{\sqrt{2}}$$

$$0.005 - 0.005 = 0$$

$$0.005 - 0.005 = 0$$

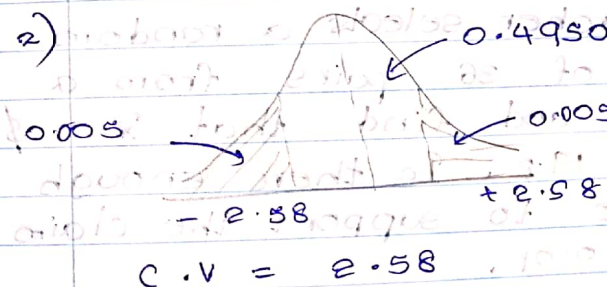
$$0.005 - 0.005 = 0$$

A medical foundation reports that the average cost of a cardiac bypass is \$24,672. To see the average cost of a bypass, a researcher select a random sample of 35 patients and finds that the average cost to be \$25,226. The σ is \$3251. At $\alpha = 0.01$, can it be concluded that the average cost of a bypass is different from \$24,672.

$$\mu = \$24,672 \quad n = 35 \quad \alpha = 0.01$$

$$\sigma = 3251 \quad \bar{x} = \$25,226$$

- 1) $H_0: \mu = 24,672$ population A
 $H_1: \mu \neq 24,672$ population B



$$z = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}} = \frac{25.226 - 24.672}{19.2 / \sqrt{36}} = 1.0100$$

4) Do not reject the null

5) There is not enough evidence to support the claim that the average cost of sugary is different from \$24.672.

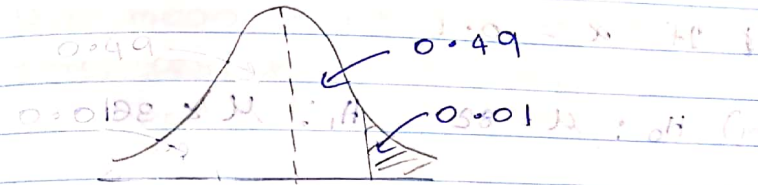
A marketing Journal claims that the average cost of a pair of athletic shoes is less than \$80. A researcher selects a random sample of 36 pairs from a catalog and find that $\bar{X} = \$75$, $\sigma = \$19.2$. Is there enough evidence to support the claim at $\alpha = 0.01$.

right tail or left tail? Since $\mu = 80$ and $\bar{X} = 75$, it is left tail. Because it is symmetric.

$$\mu = 80, \sigma = 19.2, n = 36, \bar{X} = 75, \alpha = 0.01$$

$$H_0: \mu = 80 \quad H_1: \mu < 80$$

2)



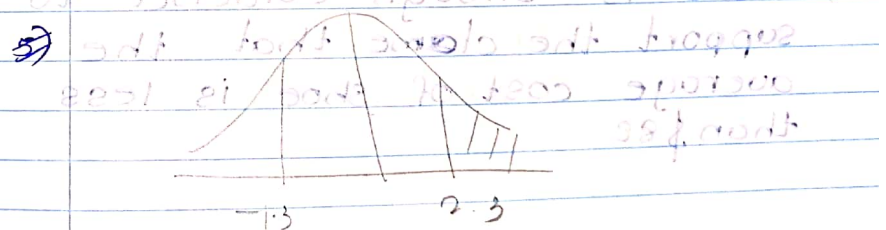
$$C.V. = -2.33$$

$$z = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}} = \frac{75 - 80}{19.2 / \sqrt{36}} = -\frac{5}{3.2} = -1.56$$

Decision: $-1.56 > -2.33$, Do not reject the null.

4) Reject the null

There is enough evidence to support the claim that the average cost of athletic shoes is less than \$80.

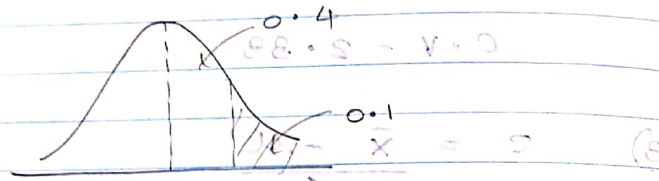


3) There is no evidence to support the claim that the average cost of shoes is less than 80.

If $\alpha = 0.1$

01) $H_0: \mu = 80$ $H_1: \mu < 80$

02)



C.V = 78.2

03) $z = -1.58$

04) Since -1.58 falls in the critical region, we reject the null hypothesis.

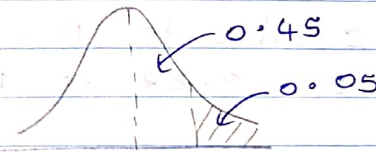
05) There is enough evidence to support the claim that the average cost of shoe is less than \$80.

A researcher wishes to test the claim that the average cost of attending university is greater than \$5700. The researcher selects 36 universities and finds the mean to be \$5950. $\sigma = \$659$. Is there enough evidence to support the claim at $\alpha = 0.05$.

$\mu = 5700$ $n = 36$ $\bar{x} = 59500$
 $\sigma = 659$ $\alpha = 0.05$

01) $H_0: \mu = 5700$ $H_1: \mu > 5700$

02)



C.V $\rightarrow 5750$

03) $z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$
 $= \frac{59500 - 57500}{659/\sqrt{36}}$
 $= \frac{200}{659/6}$

$z = 1.82$

04) reject null, because test statistic is in the critical region.

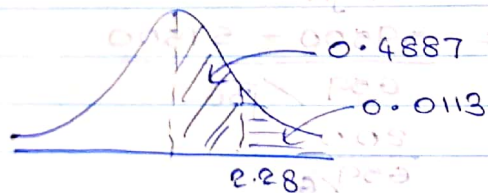
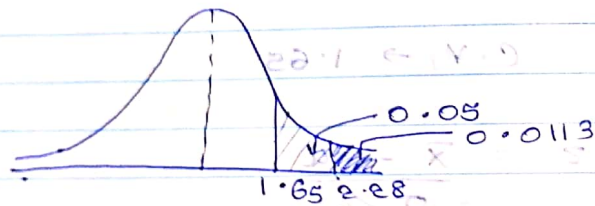
05) There is enough evidence to support the claim that tuition fees is greater than \$5700.

P Value method.

01. State hypotheses.

02. Computing test statistic (z value)

03. Convert z value to P value.
Ex - $z = 2.28 \rightarrow P = 0.4887$



P value of our test statistic is less than α

Reject null because p value corresponding to test statistic is less than α .

If you use p value method, p of test statistic $\leq \alpha$ - Reject null.

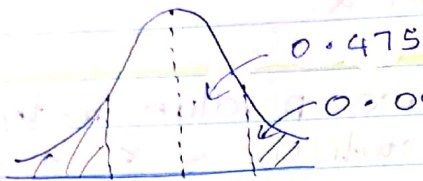
If p of test statistic $> \alpha$ - Fail to reject null.

A researcher claim that the average wind speed in a certain city is 8 miles per hour. A sample of 32 days has an average speed of 8.2 miles per hour. The standard deviation of the population is 0.6 mph. at $\alpha = 0.05$ is there enough evidence to reject the claim. Use both traditional and p value method.

$$\mu = 8 \quad n = 32 \quad \bar{x} = 8.2$$
$$\sigma = 0.6 \quad \alpha = 0.05$$

1) $H_0: \mu = 8$, $H_1: \mu \neq 8$

2)



c.v = ± 1.96

3) $z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$
 $= \frac{8.2 - 8}{0.6/\sqrt{32}}$

4) Do not reject the null. (since test statistic is outside the critical region)

5) There is not enough evidence to support the claim that the average wind speed is 8 mph

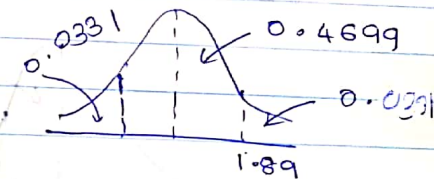
Pvalue method

01) $H_0: \mu = 8$, $H_1: \mu \neq 8$

02) $z = 1.88$

03) $P = 0.0362$

04) $\alpha = 0.05$



$P < 0.05$ $P > 0.05$

Do not reject the null.

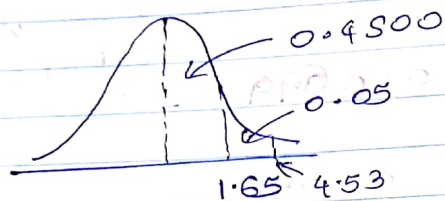
05)

If claim says that the,
 * value can be different from $\mu \rightarrow$

* Value is greater than $\mu \rightarrow$
 right tail

* Value is less than $\mu \rightarrow$
 Left tail

$\mu = 5000$ $\bar{x} = 5480$ $\alpha = 0.05$
 $n = 40$ $\sigma = 600$
 $H_0: \mu = 5000$ $H_1: \mu > 5000$



$C.V = 1.65$

$$\begin{aligned}
 z &= \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} \\
 &= \frac{5480 - 5000}{600 / \sqrt{40}} \\
 &= \frac{480}{600 / \sqrt{40}} \\
 &= 0.716 \times \sqrt{40} \\
 &= 4.53
 \end{aligned}$$

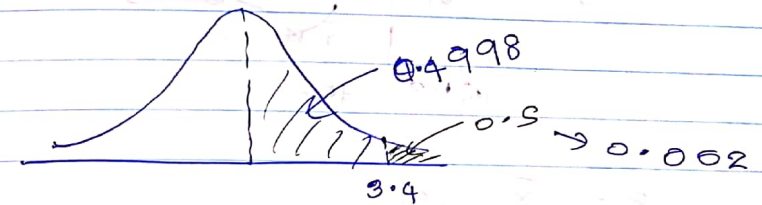
~~reject the null~~
 reject the null as $z = 4.53$
 in critical region.

~~There is not enough evidence to support the claim that average walking speed is 5480 steps greater than 5000.~~

There are enough evidence to support the claim that they walk than the average number of 5000.

P method.

$H_0: \mu = 5000$, $H_1: \mu > 5000$
 $z = 4.53$



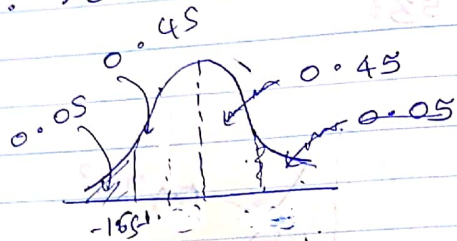
$z \rightarrow 3.4$, $P = 0.4998$

$0.0002 < 0.05$

reject null.

or) $\mu = \$3263$ $\bar{x} = \$2995$ $\alpha = 0.05$
 $n = 50$ $\sigma = \$1100$

$H_0: \mu = 3263$; $H_1: \mu < 3263$



$$z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

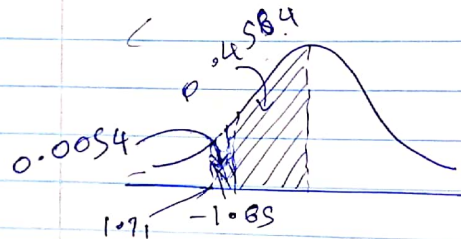
$$= \frac{2995 - 3263}{1100 / \sqrt{50}}$$

$$= \frac{-268}{155.56}$$

$$= -1.71$$

P method

$z = -1.71$ $z = -1.71$



$P = 0.4584$

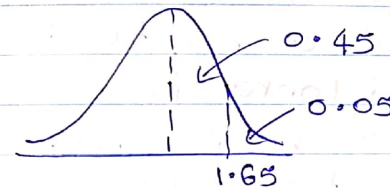
$0.05 < 0.4584$
 $0.05 > 0.0064$

Reject the null

There are enough evidences to support the claim that seniors have a debt much less than \$3263.

or) $\mu = \$12837$ $\bar{x} = \$14445$
 $n = 44$ $\sigma = \$1500$
 $\alpha = 0.05$

$H_1: \mu = 12837$ $H_0: \mu > 12837$



$C.V = 1.65$

$$z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

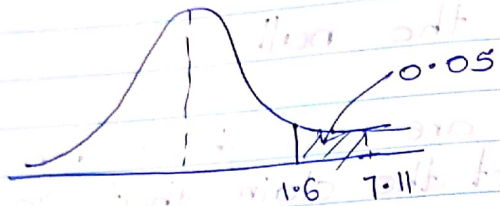
$$= \frac{14445 - 12837}{1500 / \sqrt{44}}$$

$$= \frac{1608}{226.13}$$

$$= 7.1$$

P method.

$$z = 7.11$$



Reject null

There are enough evidence to support the claim that se PhD student earns more than \$12837

A medical investigation claims that the average number of infections per week at a hospital in south west of Sri Lanka is 16.3. A random sample of 10 weeks had a mean of 17.7 infections. The sample standard deviation is 1.8. Is there enough evidence to reject the investigator claim. $\alpha = 0.05$

$$\mu = 16.3 \quad \bar{x} = 17.7 \quad \alpha = 0.05$$

$$n = 10 \quad s = 1.8$$

$$H_0: \mu = 16.3$$

$$H_1: \mu \neq 16.3$$

$$df = n - 1$$

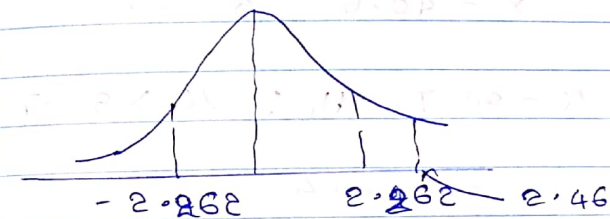
$$= 9$$

$$c.v = 2.262$$

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

$$= \frac{17.7 - 16.3}{1.8/\sqrt{10}}$$

$$= 2.46$$



Reject null

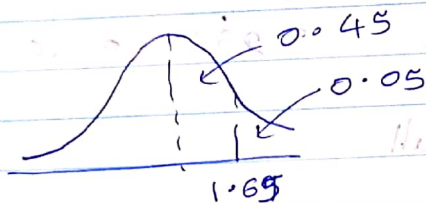
There is enough evidence to reject the claim that the average number of infections is 16.5.

A physician claims that an athlete's maximum oxygen uptake is greater than the average adult. A sample of 15 joggers has a mean of 40.6 ml/kg. The standard deviation is 6 ml/kg. If the average of all adults is 36.7 ml/kg. Is there enough evidence to support the physician's claim at $\alpha = 0.05$.

$$\mu = 36.7 \quad \alpha = 0.05$$

$$n = 15 \quad \bar{x} = 40.6 \quad s = 6$$

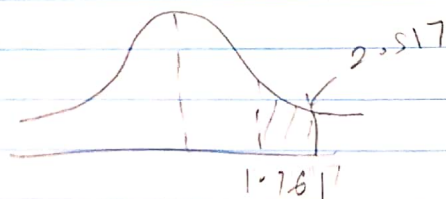
$$H_0 : \mu = 36.7 \quad H_1 : \mu > 36.7$$



$$C.V = 1.65$$

$$Z = C.V = 1.65$$

$$t = 2.517$$



Test the differences between two mean.

What is the difference in age between students at RUSL and CMB

$$RUSL - \mu_1$$

$$CMB - \mu_2$$

μ_1 - mean age of students of RUSL
 μ_2 - mean age of students of CMB

$$H_0 : \mu_1 = \mu_2 \rightarrow \mu_1 - \mu_2 = 0$$

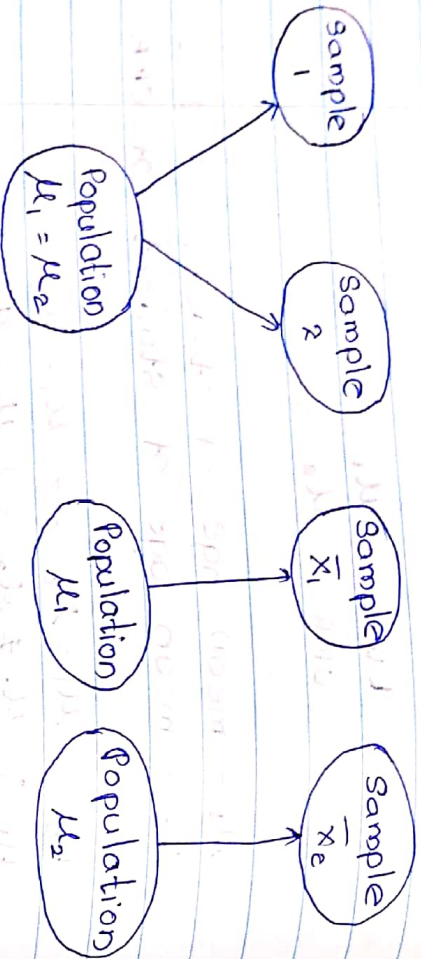
$$H_1 : \mu_1 \neq \mu_2 \rightarrow \mu_1 - \mu_2 \neq 0$$

Assumptions,

- The sample should be independent of each other.
- The standard deviation of both population should be known. for sample size less than 30, normally distributed.

$$z = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

$$\mu_1 - \mu_2 = 0$$



$$H_0: \mu_1 = \mu_2$$

The difference is not significant.

Fail to reject null

$$H_0: \mu_1 = \mu_2$$

The difference is significant.

Reject null

A survey found that the average hotel rate in Kandy is \$ 88.42 and the average in Anuradhapura is \$ 80.61. Assume that the data was obtained from two samples of 50 each. The standard deviation of the population are \$ 5.62 and \$ 4.83 respectively. At $\alpha = 0.05$

can it be concluded that there is a significant difference in the rates.

$$H_0: \mu_1 = \mu_2, \mu_1 - \mu_2 = 0$$

$$H_1: \mu_1 \neq \mu_2, \mu_1 - \mu_2 \neq 0$$

two tail test.

$$\alpha = 0.05 \quad C.V = \pm 1.96$$

03)

$$z = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

$$= \frac{(88.42 - 80.61) - 0}{\sqrt{\frac{5.62^2}{50} + \frac{4.83^2}{50}}}$$

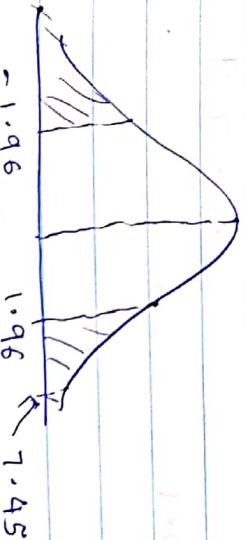
$$= \frac{7.81}{\sqrt{\frac{31.58}{50} - \frac{23.3}{50}}}$$

$$= \frac{7.81}{\sqrt{1.098}}$$

$$= 7.45$$

$$= 7.81$$

$$= 7.45$$



04) Reject the null.

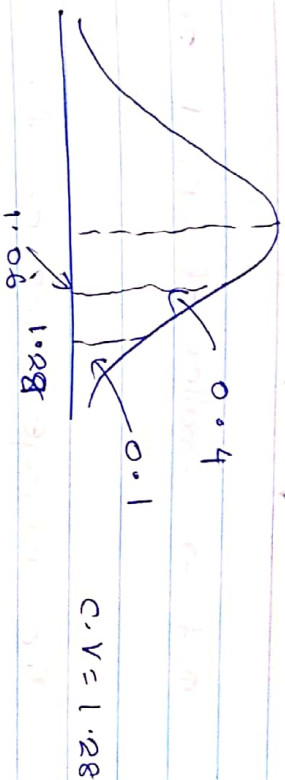
05) There is enough evidence to support the claim that the means are not enough equal.

A researcher hypothesizes that the average number of sports that universities offer for males is greater than the average number of sports for females. A sample of 50 universities was considered. The average number of sports offered to males was 8.6 with a population standard deviation of 3.3, while the average number of sports offered to female was 7.9 with $\sigma = 3.3$. $\alpha = 0.1$. Is there enough evidence to support the claim.

$n = 50$
 $\alpha = 0.1$
 $H_0: \mu_1 = \mu_2$
 $H_a: \mu_1 > \mu_2$
 $\bar{x}_1 = 8.6$
 $\bar{x}_2 = 7.9$
 $\sigma_1 = 3.3$
 $\sigma_2 = 3.3$

01) $H_0: \mu_1 = \mu_2$, $\mu_1 - \mu_2 = 0$
 $H_1: \mu_1 > \mu_2$, $\mu_1 - \mu_2 \neq 0$

02) C.V



$$z = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n} + \frac{\sigma_2^2}{n}}}$$

$$z = \frac{(8.6 - 7.9) - 0}{\sqrt{\frac{3.3^2}{50} + \frac{3.3^2}{50}}}$$

$$z = \frac{0.7}{\sqrt{0.4356}}$$

$$z = \frac{0.7}{0.66}$$

$$z = 1.06$$

04) Fail to reject null

05) There is not enough evidence to support the claim

When σ is not known.

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}$$

D.f $\rightarrow$ smaller of $n_1 - 1$ or $n_2 - 1$

The average size of farm in Anuradhapura is 191 acres. The average size of a farm in Polonnaruwa is 199 acres. Assume this data was obtained from two samples with standard deviation of 38 and 12 acres respectively. and sample size of 8 and 10 can it be concluded at $\alpha = 0.05$ that the average size of farms in the two districts is different? Assume that the population is normally distributed.

sample size is less than 30

A
 $\bar{x} = 191$ $x = 199$
 $\sigma = 38$ $\sigma = 12$ $\alpha = 0.05$
 $n = 8$ $n = 8$

01) $H_0: \mu_1 = \mu_2$ $\mu_1 - \mu_2 = 0$

$H_1: \mu_1 \neq \mu_2$ $\mu_1 - \mu_2 \neq 0$

02) two tail $\alpha = 0.05$

ev $= 1.96$ d.f - $8 - 1 = 7$

$10 - 1 = 9$



* take smaller degree of freedom.

C.V. $= \pm 2.865$

03) $t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}$

$$t = -0.575$$

04) Fail to reject the null

A bank spent an average of 21 minutes helping each of 10 people. The standard deviation was 5.6 minutes. A second bank spend 27 minutes on average helping 14 people. The standard deviation was 4.8 min. At $\alpha = 0.02$ is there a different in the average time spent by the two banks on their customers.

	$\bar{x}$	n
A	21	10
B	27	14

$$\alpha = 0.02$$

$$H_0: \mu_1 = \mu_2 \quad \mu_1 - \mu_2 = 0$$

$$H_1: \mu_1 \neq \mu_2 \quad \mu_1 - \mu_2 \neq 0$$

two tail

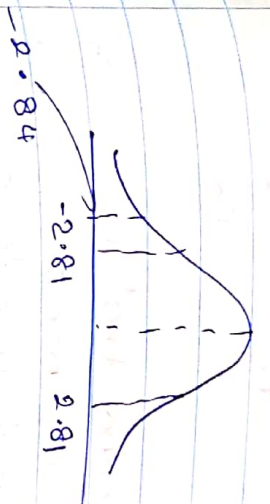
$$n-1 = 9$$

$$C.V = \pm 2.821$$

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

$$= \frac{(21 - 27) - 0}{\sqrt{\frac{5.6^2}{10} + \frac{4.8^2}{14}}} = -2.84$$

04)



Reject null.

05) have enough evidence.

Testing for difference between proportions.

P_1

P_2

$$H_0: P_1 = P_2$$

$$H_1: P_1 \neq P_2$$

$$P_1 - P_2 = 0$$

$$P_1 - P_2 \neq 0$$

z test for proportions.

$$z = \frac{(\hat{P}_1 - \hat{P}_2) - (P_1 - P_2)}{\sqrt{\bar{P}\bar{Q} \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

$$\bar{P} = \frac{X_1 + X_2}{n_1 + n_2} \quad \bar{Q} = 1 - \bar{P}$$

$$\hat{p}_1 = \frac{x_1}{n_1}$$

$$\hat{p}_2 = \frac{x_2}{n_2}$$

Researches found that 12 out of 34 small nursing homes had a resident vaccination rate of less than 80%, while 17 out of 24 large nursing homes had a vaccination rate of less than 80%. at $\alpha = 0.05$ test the claim that there is no difference in the proportion of the small and large nursing homes with regards to resident vaccination rate.

$$01) H_0: p_1 = p_2$$

$$H_1: p_1 \neq p_2$$

$$02) \alpha = 0.05 \quad \text{two tail}$$

$$CV = \pm 1.96$$

$$\hat{p} = \frac{x_1}{n_1} = \frac{12}{34} = 0.35$$

$$\hat{p}_2 = \frac{x_2}{n_2} = \frac{17}{24} = 0.71$$

$$\bar{p} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{12 + 17}{34 + 24}$$

$$= 0.5$$

$$\bar{q} = 0.5$$

$$z = \frac{(\hat{p}_1 - \hat{p}_2) - (\bar{p}_1 - \bar{p}_2)}{\sqrt{\bar{p}\bar{q}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

$$= \frac{(0.35 - 0.71) - 0}{\sqrt{(0.5)(0.5)\left(\frac{1}{34} + \frac{1}{24}\right)}}$$

$$= -2.65$$

04) Reject

Chi-square test

Deals with frequencies.

Two types of chi-square tests

01. Chi-square - Goodness of Fit (GOF)

For this we use chi square distribution.

- not symmetrical
- shape is depend on the sample size / degrees of freedom

	Cherry	straw	orange	lime	grape
observed	28	28	16	14	10
Expected	20	20	20	20	20

01. H_0 - There is no preference for flavours among respondents.
 H_1 - Respondents show a preference.

02.

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

 E - expected value (frequency)
 O - observed value (frequency)

03. Find critical value.

$$d.f = n - 1 = 5 - 1 = 4 //$$

$$C.V \quad \alpha = 0.05 \quad C.V = 9.488$$

χ^2 - table

* Assumptions

- 1) The data has been obtained from random sample.
- 2) Expected frequency in each category should be greater than 5.

$$\begin{aligned} \chi^2 &= \frac{(28-20)^2}{20} + \frac{(28-20)^2}{20} + \frac{(16-20)^2}{20} \\ &+ \frac{(14-20)^2}{20} + \frac{(10-20)^2}{20} \\ &= 18 \end{aligned}$$



09) reject the null.
Because test statistic is in the critical region.

03) There is enough evidence to reject the claim that consumers/respondents show preference for flavors.

A survey was conducted among retired executives. They found that 80% were employed somewhere else, 32% self employed, 23% were consulting, 7% had formed their own companies. To see if these percentages were are consistent with the above survey a local researcher surveys 300 retired executives, and found 112 were working for another

company, 85 self employed, 76 consulting and 17 formed their own companies. At $\alpha = 0.10$, test the claim that the two percentages are the same.

01. H_0 - The retired executives are distributed as follows,

88% - employed somewhere else
32% - self employed
23% - consulting
7% - form their own companies

H_1 - The distribution not the same as distributed in the null.

02. d.f = 4 - 1 = 3

C.V = 6.251

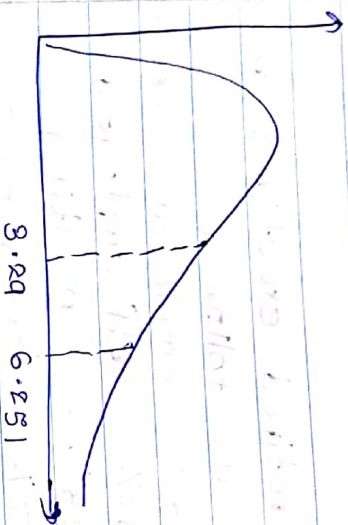
03 Expected value for our sample;

$0.38 \times 300 = 114$ $0.23 \times 300 = 69$
 $0.32 \times 300 = 96$ $0.07 \times 300 = 21$

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

$$= \frac{(116 - 114)^2}{114} + \frac{(85 - 96)^2}{96} + \frac{(76 - 69)^2}{69} + \frac{(17 - 21)^2}{21}$$

$$= 8.9$$



04. Fail to reject the null.

05. There is not enough evidence to reject the claim. It can be concluded that the percentages differ significantly from those given in the null.

Chi-Square test of independence.

Preference for medical procedure by doctors and nurses.

Groups	Prefer new	Prefer old	Preference
Nurse	100	80	20
Doctor	50	120	30

Is opinion linked to profession,

01. H_0 :- The opinion about the procedure is independent of profession,

H_1 - The opinion about procedure is dependent of profession.

$$\begin{aligned} \text{d.f.} &= (R-1)(C-1) & R - \text{rows} \\ &= (2-1)(3-1) & C - \text{columns} \\ &= 2 \end{aligned}$$

$$\alpha = 0.05$$

$$C.V. = 5.991$$

$$\text{Expected Value} = \frac{\text{row sum} \times \text{column sum}}{\text{grand total}}$$

Contingency table

Group	Prof new	Prof old	No Pre	Total
Nurse	100	80	20	200
Doctor	50	120	30	200
	150	200	50	400

Group	Prof new	Prof old	No Pre	Total
Nurse	100 (75)	80 (100)	20 (25)	200
Doctor	50 (75)	120 (100)	30 (25)	200
	150	200	50	400

$$E_{(1,1)} = \frac{200 \times 150}{400} = 75$$

$$E_{(1,2)} = \frac{200 \times 200}{400} = 100$$

$$E_{(1,3)} = \frac{200 \times 50}{400} = 25$$

$$E_{(2,1)} = \frac{200 \times 150}{400} = 75$$

$$E_{(2,2)} = \frac{200 \times 200}{400} = 100$$

$$E_{(2,3)} = \frac{200 \times 50}{400} = 25$$

$$\chi^2 = \sum \frac{(O - E)^2}{E} = \frac{(100 - 75)^2}{75} + \frac{(80 - 100)^2}{100} + \frac{(20 - 25)^2}{25} + \frac{(50 - 75)^2}{150} + \frac{(120 - 100)^2}{200} + \frac{(30 - 25)^2}{50}$$

$$= \frac{(20 - 25)^2}{25} + \frac{(30 - 25)^2}{25} + \frac{(80 - 100)^2}{100} + \frac{(120 - 100)^2}{200} + \frac{(20 - 25)^2}{25} + \frac{(30 - 25)^2}{25}$$

$$= 25.16$$



Reject null, there is

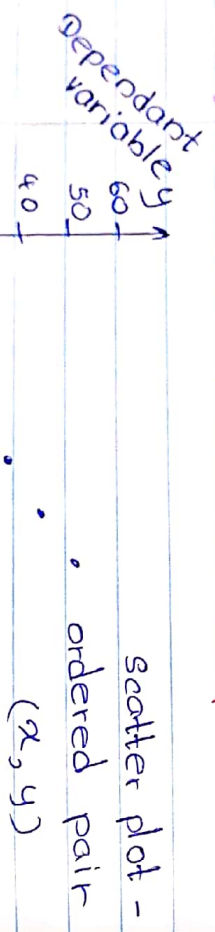
There is enough evidence to support the claim that opinion is associated with profession,

- * Is there a relationship between environmental pollution and lung diseases.
- * Is there a relationship between the number of forests per hectare in forested areas and bird diversity.

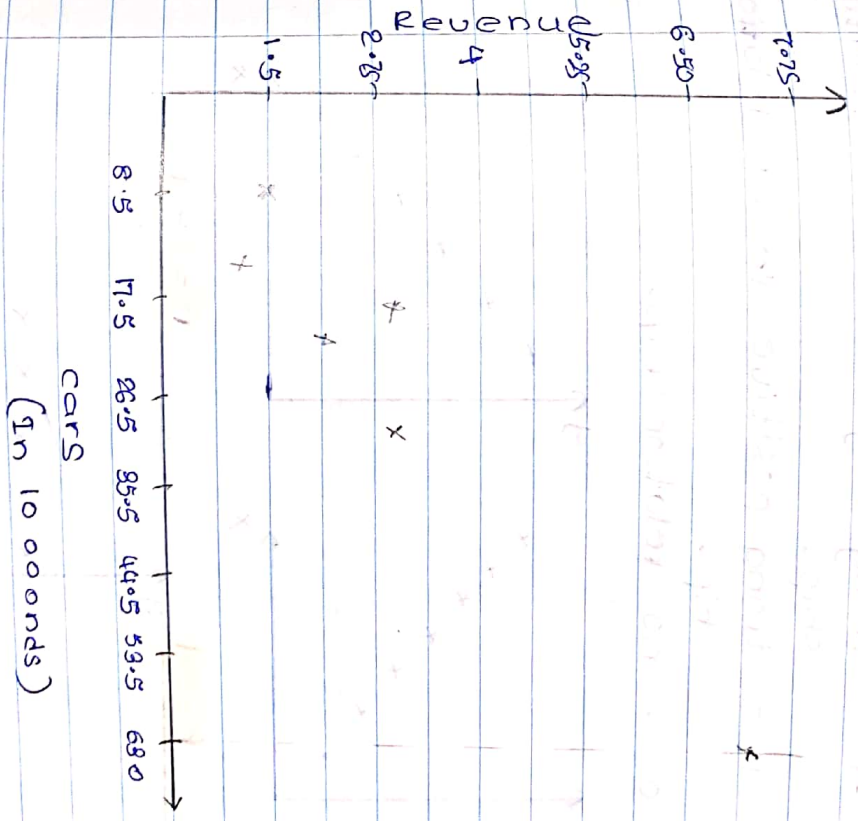
We can ask 4 types of questions

- 01) Are the two variables related?
- 02) If so what is the strength of the relationship?
- 03) What type of relationship exists?
- 04) What kind of prediction can be made from the relationship?

01) Is there a relationship?



company	cars (in thousands)	revenue (in billions)
A	63.0	7.0
B	29.0	2.9
C	20.8	2.1
D	19.1	2.8
E	13.4	1.4
F	8.5	1.5



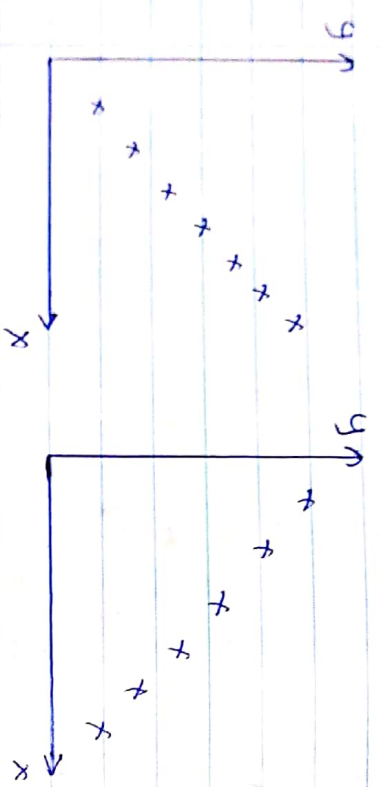
$r = 1$

Correlation coefficient

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n(\sum x^2) - (\sum x)^2][n(\sum y^2) - (\sum y)^2]}}$$

$r = -1$ to $+1$

- 1 = strong negative linear relation -ship.
- +1 = strong positive linear relation -ship.
- 0 = no relationship.



$r = 1$



$r = 0$

$r = -1$

cars (x)	Income (y)	xy	x ²	y ²
----------	------------	----	----------------	----------------

A	63.0	7.0	441	3969	49
B	29.0	3.9	113.1	841	15.21
C	20.8	2.1	43.68	432.64	4.41
D	19.1	2.8	53.48	364.81	7.84
E	13.4	1.4	18.76	179.56	1.96
F	8.5	1.5	12.75	72.25	2.25
$\sum$	153.8	18.7	682.77	5859.26	80.67

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n(\sum x^2) - (\sum x)^2][n(\sum y^2) - (\sum y)^2]}}$$

$$= \frac{6(682.77) - (153.8)(18.7)}{\sqrt{[6(5859.26) - (153.8)^2][6(80.67) - (18.7)^2]}}$$

$$= 0.97$$

There is a strong positive relation -ship.

No: _____

Date: ____/____/____

Number of hours of exercise and milk consumption.

Subject	Hourse. Amount (x)	(y)	xy	x ²	y ²	
A	3	48	114	9	2304	
B	0	8	0	0	64	
C	2	32	64	4	1024	
D	5	64	320	25	4096	
E	8	10	80	64	100	
F	5	32	160	25	1024	
G	10	56	560	100	3134	
H	2	72	144	4	5184	
I	1	48	48	1	2304	
Σx = 36			Σy = 370	Σxy = 1520	Σx ² = 232	Σy ² = 1924

$$r = \frac{n(\Sigma xy) - (\Sigma x)(\Sigma y)}{\sqrt{[n(\Sigma x^2) - (\Sigma x)^2][n(\Sigma y^2) - (\Sigma y)^2]}}$$

$$= \frac{9(1520) - (36)(370)}{\sqrt{[9(232) - (36)^2][9(1924) - (370)^2]}}$$

$$= 0.067$$

Weak positive relationship.

If,

$r = 0.97 \approx 1$
 $r = 0.76, 0.68 \rightarrow$ If there is a relationship or not, not sure.

No: _____

Date: ____/____/____

Significance or hypothesis test for r , whether population correlation coefficient ρ ,

$H_0, \rho = 0$ - there is no relationship between two variables
 $H_1, \rho \neq 0$ - there is a relationship between variables.

- 01) state the hypothesis.
- 02) Find the critical value.
- 03) Complete test statistic.
- 04) Make a decision.
- 05) Summarizing result.

$$t = r \sqrt{\frac{n-2}{1-r^2}}$$

Degrees of freedom D.F = $n-2$

from 1st example,

car sales vs Income $r = 0.97$.

$$D.F = 6 - 2 = 4$$

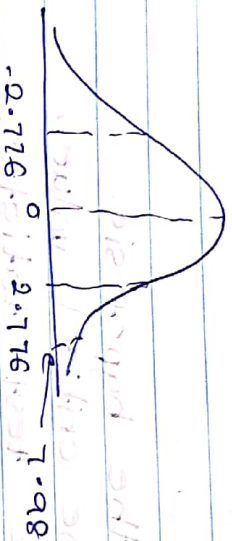
$$\alpha = 0.05$$

oe. two tail test.

$$C.V = \pm 2.776$$

$$t = 0.97 \sqrt{\frac{6-2}{1-(0.97)^2}}$$

$$= 2.798$$



Reject the null hypothesis.

There is a significant relationship between the number of cars sold and the income.

from 2nd example.

$$H_0, \rho = 0 \quad H_1, \rho \neq 0, \alpha = 0.05$$

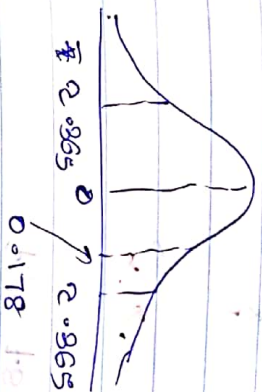
$$D.F = 9-2$$

$$= 7$$

$$C.V = \pm 2.365$$

$$t = r \sqrt{\frac{n-2}{1-r^2}}$$

$$= 0.067 \sqrt{\frac{9-2}{1-(0.067)^2}} = 0.178$$



Fail to reject the hypothesis.

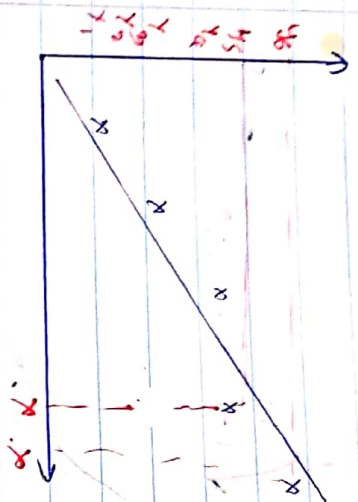
There is not enough evidence to support the claim that there is a relationship between the two variables.

Assumptions

- * The variable x and y are linearly related.
- * The variables are random variables.
- * The two variables have a bivariate normal distribution.

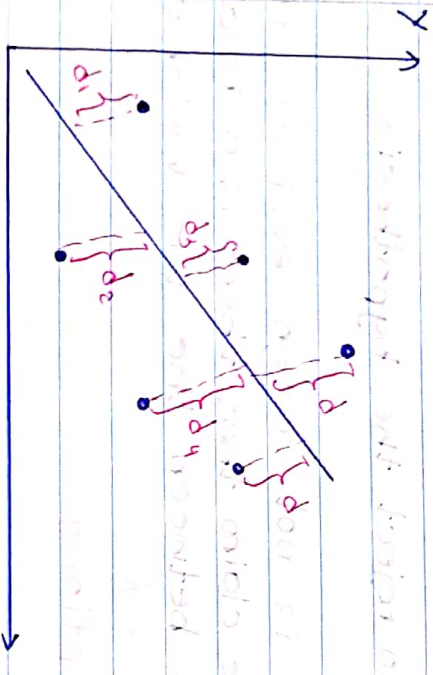
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$$r = 0.97$$

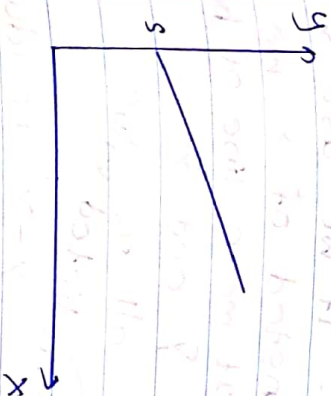
Line of best fit



The line that minimises the vertical distance between the observed and the line (predicted values)

No: _____

Date: ____/____/____



for regression,

$$y' = a + bx$$

$$a = \frac{(\sum y)(\sum x^2) - (\sum x)(\sum xy)}{n(\sum x^2) - (\sum x)^2}$$

$$b = \frac{n(\sum xy) - (\sum x)(\sum y)}{n(\sum x^2) - (\sum x)^2}$$

From 1st example $\rightarrow$ car sale,

30/07/2021

$$a = 0.896 \quad b = 0.106$$

$$y' = 0.396 + 0.106x \quad \bar{Y} - Y \quad Y$$

$$x = 15$$

$$y' = 0.396 + 0.106 \times 15$$

$$= 1.986$$

We can use for x values 10 and 60 (range of the given data)

We can only use the line to predict within this range.

For best fit, Predicted value (y') close to observed value (y), and Residual very low.

Graph showing - If we use $\bar{x}$ in y' equation instead of x we should get $\bar{y}$. It means line of best fit passes through $\bar{y}$ and $\bar{x}$. Calculate y' for all x value

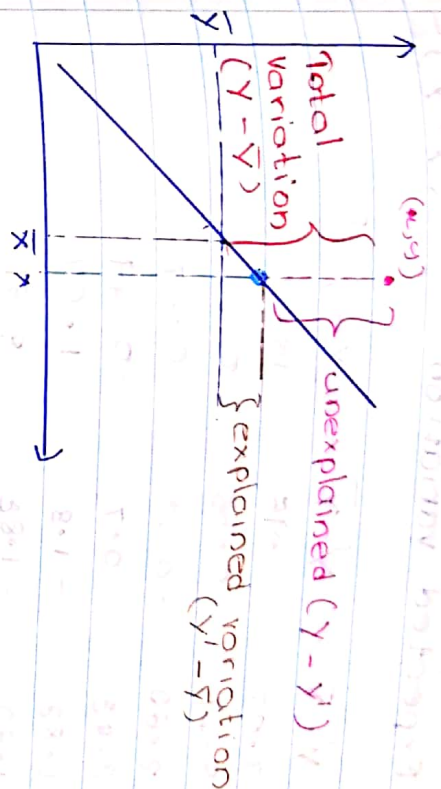
x	y'	y	$y - y'$ (residual)
6.5	7.07	7	-0.07
6.9	8.47	8.9	0.43
6.8	8.60	2.1	-0.51
19.1	2.42	2.8	0.38
18.4	1.82	1.4	-0.42
8.5	1.90	1.5	0.2
	18.7	0	0

* smaller difference better. ($-y$) at predicting

Total variation of $y = \sum (y - \bar{y})^2$

y	$y - \bar{y}$	$(y - \bar{y})^2$
3.9	0.78	0.64
7	3.88	15.05
2.1	-1.02	1.04
2.8	-0.32	0.102
1.4	-1.72	2.95
1.5	-1.62	2.62

$$\bar{y} \Rightarrow \text{mean} = \frac{18.7}{6} = 3.12$$



Total variation = Explained + unexplained deviation

Unexplained variation $\sum (y - y')^2$

$y - y'$	$(y - y')^2$
-0.07	0.0049
0.43	0.1849
-0.51	0.2601
-0.38	0.1444
-0.42	0.1764
0.2	0.04
	0.81

unexplained variation

Expected variation = $\sum (y' - \bar{y})^2$

$y' - \bar{y}$	$y' - \bar{y}$	$(y' - \bar{y})^2$
7.07	3.95	15.37
8.47	0.35	0.12
2.60	-0.52	0.27
2.42	-0.7	0.49
1.82	-1.3	1.69
1.30	-1.82	3.31
		<u>21.27</u>

Total variation = Explain + Unexplain
= 21.27 + 0.81

$$21.27 + 0.81 = 22.08$$

Coefficient of determination (r^2)

$r^2 = \frac{\text{Explain variation}}{\text{total variation}}$

$$= \frac{21.27}{22.08}$$

$$= 0.95$$

95% of the variation of (income) is explained by variation of (number of car rental companies)

If r^2 higher $\rightarrow$ better.

* the model has good predictive value. Equation is good at predicting.

Standard Error of Estimate

$$S_{est} = \sqrt{\frac{\sum (y - y')^2}{n - 2}}$$

$$= \frac{0.81}{4}$$

machine Age (years) or monthly cost (y)

A	1	78
B	2	78
C	3	70
D	4	90
E	5	93
F	6	103

Calculate r , carry out significance test for the link of best fit, the monthly cost for a machine that is 4.5 years old, r^2 , S_{est}

$\Rightarrow$

In Sri Lanka, the average cost of family insurance is Rs. 4172. A random sample of 20 families indicated that they paid an amount of Rs 45600 with standard deviation of Rs 1590. At $\alpha = 0.05$, is there enough evidence that they pay more than the national average of Rs. 4172.

$$\mu = 4172 \quad \bar{x} = 45600 \quad \sigma = 1590$$

$$n = 20 \quad \alpha = 0.05 \quad (\text{more than})$$

$$H_0: \mu = 4172 \quad H_1: \mu > 4172$$

$$d.f. = 20 - 1 \quad c.v. = 1.729$$

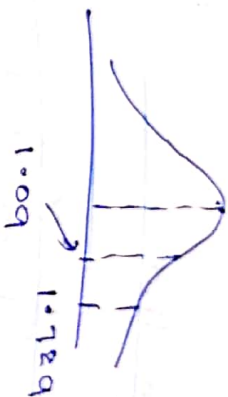
$$= 19$$

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

$$= \frac{45600 - 4172}{1590/\sqrt{20}}$$

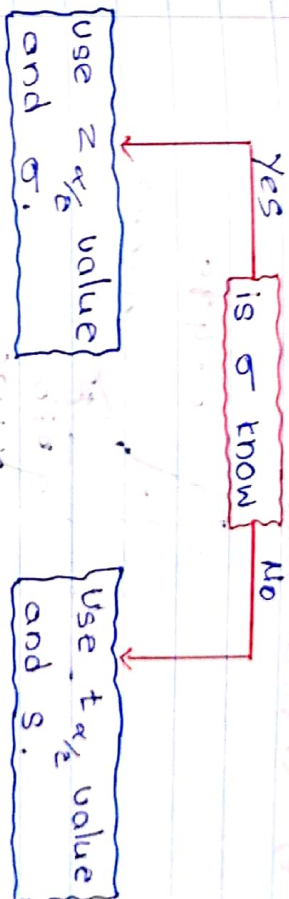
$$= \frac{41428}{355.59}$$

$$= 116.52 \quad 1.09$$



Fail to reject the null.

There is not enough evidence to support the claim that a family pays more than the national average.



* If $n \geq 30$ the variable must be normally distributed.

z test for proportion.

$$z = \frac{\hat{p} - p}{\sqrt{pq/n}}$$

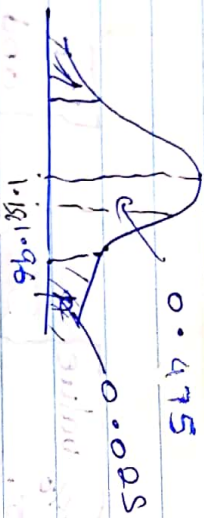
n - sample size
 $\hat{p}$ - sample proportion ($\bar{x}/n$)
 p - Population proportion

A doctor claim that 60% of people are trying to avoid trans fats.

she randomly selected 200 people and found that 128 people stated that they avoided trans fats. At $\alpha = 0.05$ is there enough evidence to reject the doctors claim.

1) $H_0: p = 0.6, H_1: p \neq 0.6$

2) critical value



3) $\hat{p} = \frac{x}{n} = \frac{128}{200} = 0.64$

$p = 0.6$

$z = \frac{\hat{p} - p}{\sqrt{pq/n}}$

$= \frac{0.64 - 0.6}{\sqrt{(0.6)(0.4)/200}} = 1.156$

4) Fail to reject.

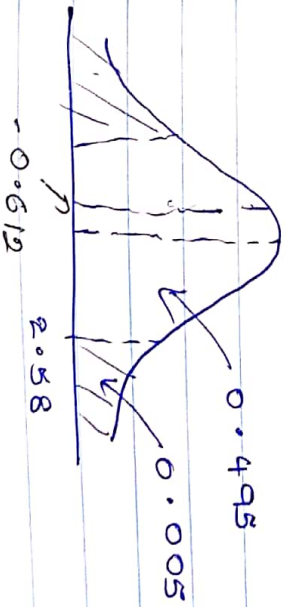
5) There is not enough evidence to reject the claim that 60% of people are trying to avoid trans fat.

A telephone company estimates that 40% of its customers have call waiting service. To test this hypothesis, a sample of 100 customers was selected and found that 37% had call waiting. At $\alpha = 0.01$, is there enough evidence to reject the claim?

$n = 100, p = 0.4, q = 0.6, \alpha = 0.01$
 $x = 37$

1) $H_0: p = 0.4, H_1: p \neq 0.4$

2)



C.V. $\approx \pm 2.58$

04/08/2021

listed below is the moisture content (by percent) for random samples of different fruits and vegetables. At the 0.05 level of significance, can it be concluded that fruits differ from vegetables in average moisture content?

Fruits sample ① Vegetables sample ②

Apricot	86	Artichoke	85
Banana	75	Bamboo shoots	91
Avocado	78	Beets	88
Blackberry	88	Broccoli	89
Clementine	87	Cucumber	95
Fig	79	Iceberg lettuce	96
Pink grapefruit	92	Mushroom	92
Mango	84	Radish	95
Tomato	94		

$$\sum (x - \bar{x})^2 = 82.875$$

$$s^2 = 47.56$$

86	3.125
75	-7.875
72	-10.875
88	5.125
87	4.125
79	-8.875
92	9.125
84	1.125

$$\sum x = 669$$

$$332.84$$

There is not enough evidence to reject the claim.

Fail to reject

$$= -0.612$$

$$\frac{\sqrt{pq/n}}{0.87 - 0.40} = \frac{\sqrt{(0.4)(0.6)/100}}{0.87 - 0.40}$$

$$p = \frac{0}{100} = 0$$